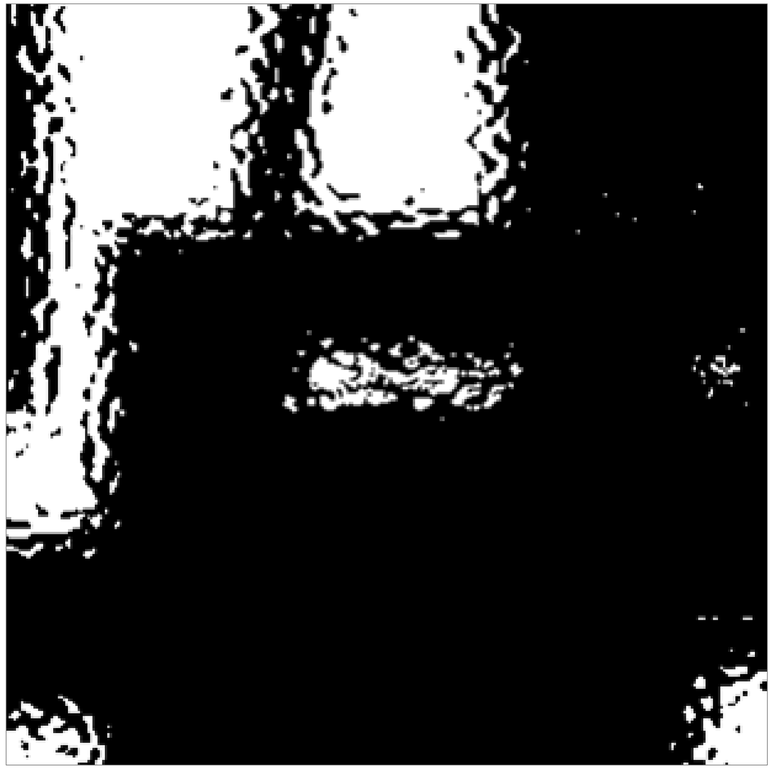
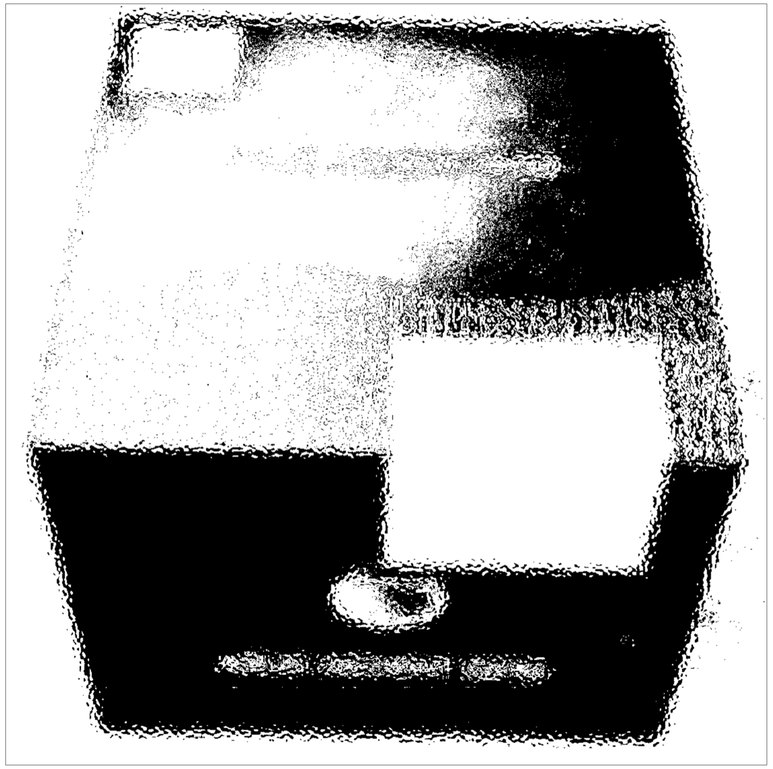
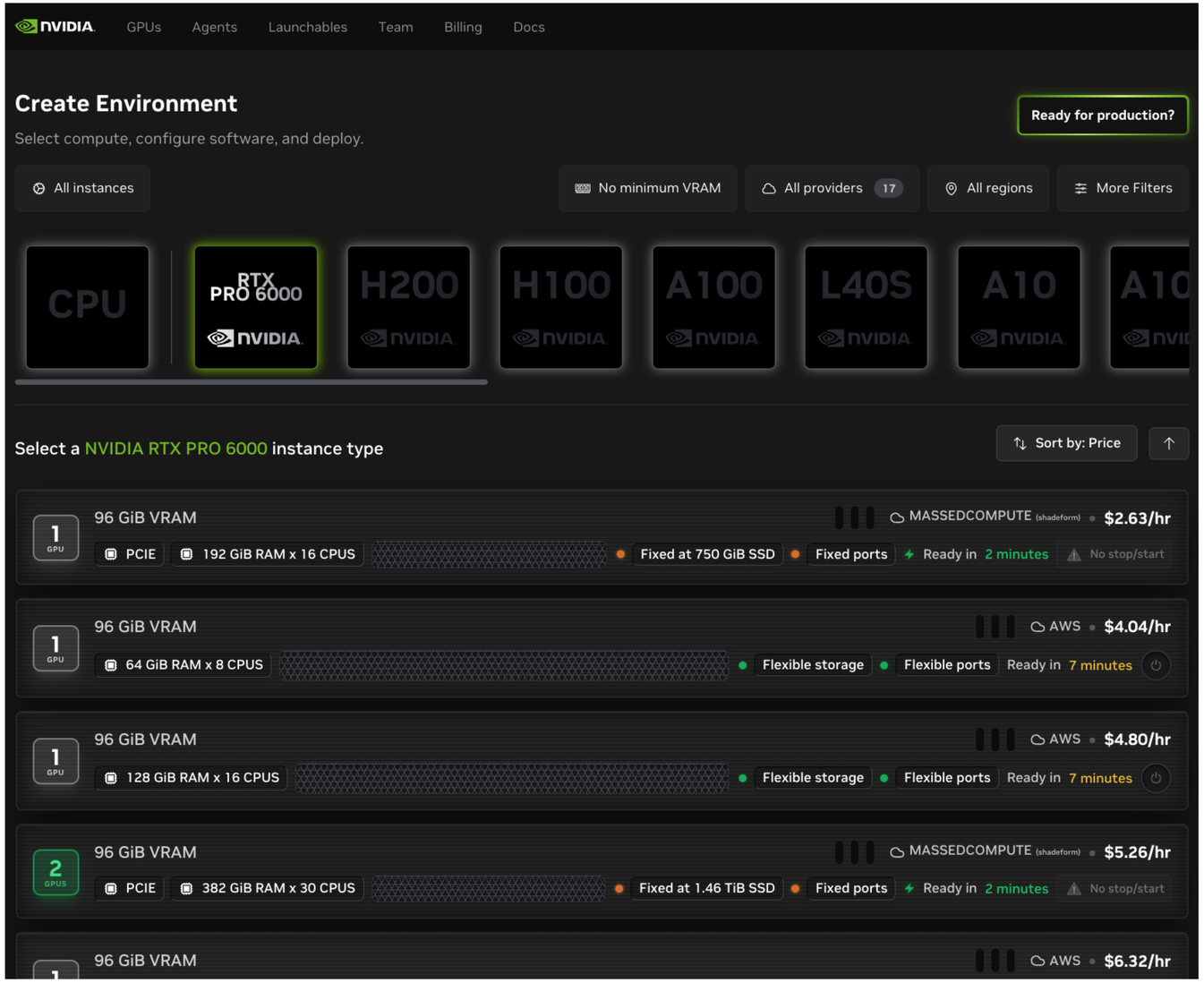
Announcing NVIDIA Special Awards for NCHC HiPAC 2026



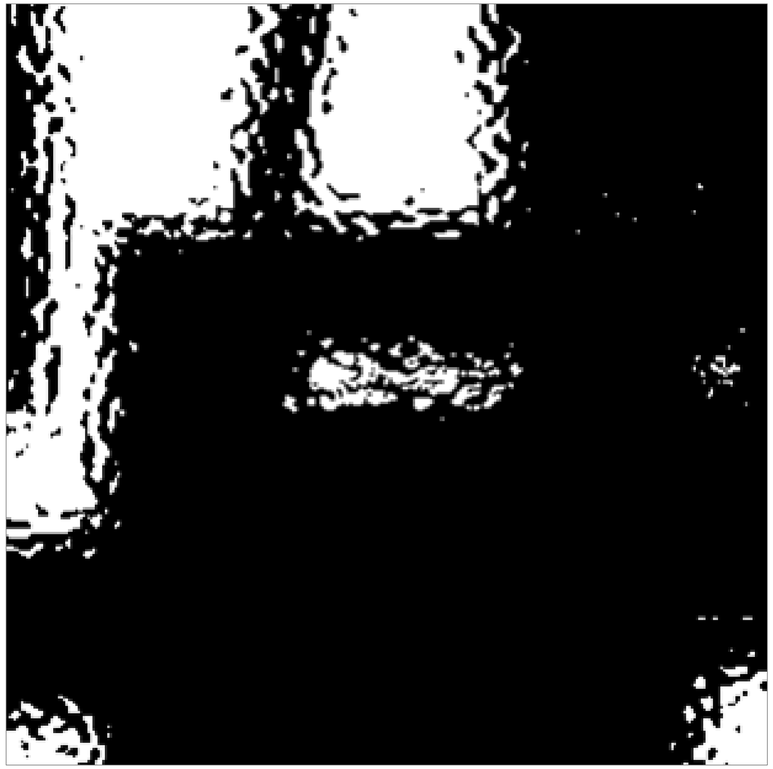


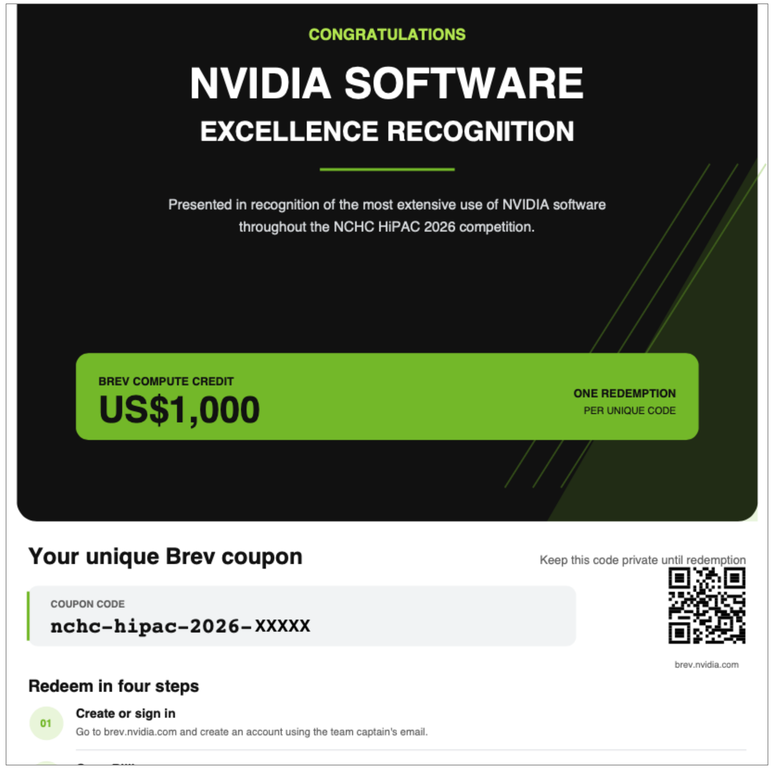
NVIDIA Special Prize: US$1,000 in NVIDIA Brev Credits
for Each of the Top Three Teams Most Aligned with NVIDIA CUDA-X

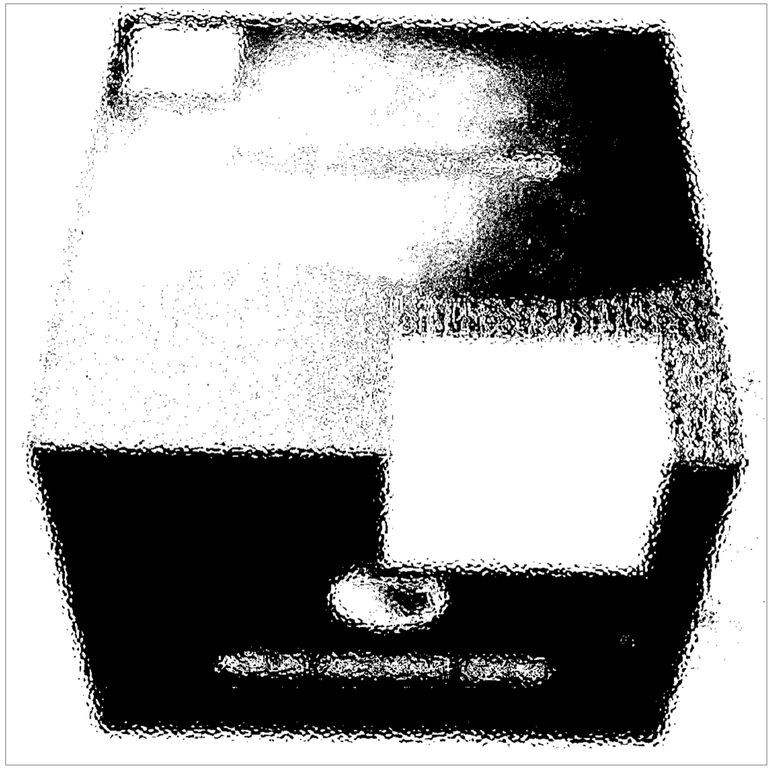
https://brev.nvidia.com/



Announcing NVIDIA Special Awards for NCHC HiPAC 2026



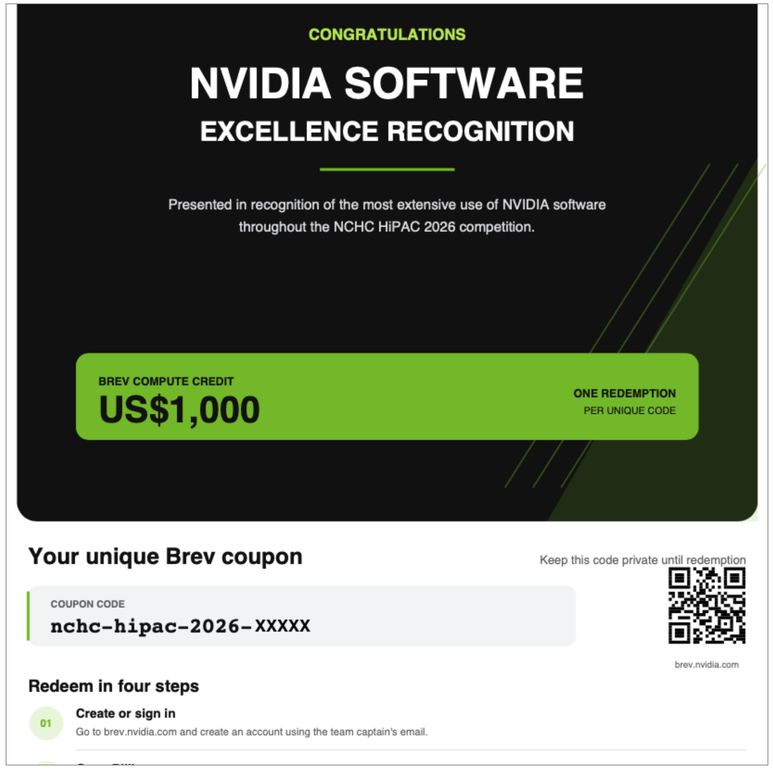




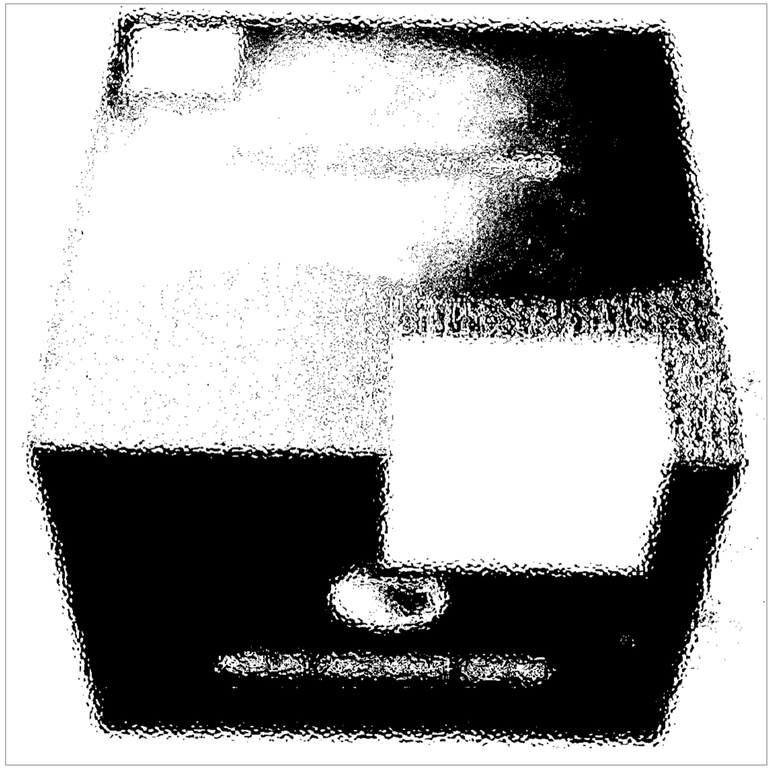
NVIDIA SOFTWARE EXELLENCE RECOGNITION FOR 3 TEAMs

Announcing NVIDIA Special Awards for NCHC HiPAC 2026

NVIDIA LOGO GTC TOTE
FOR EVERYONE



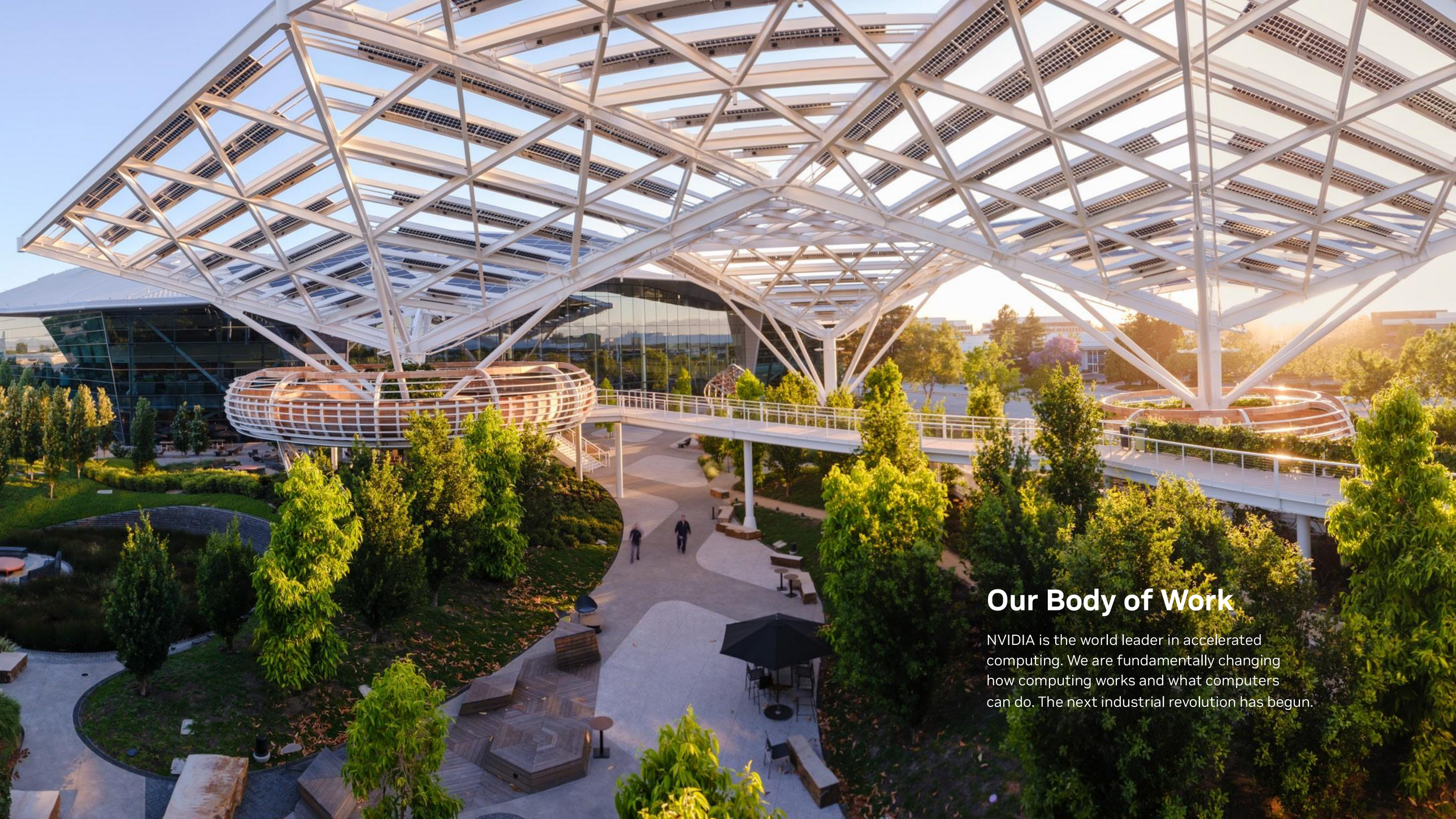
NVIDIA SOFTWARE EXELLENCE
RECOGNITION FOR 3 TEAMs



NVIDIA Platform for Science

Jay Chen, Solution Architect

NCHC HiPAC 2026 - Tech Sharing



Our Body of Work

NVIDIA is the world leader in accelerated computing. We are fundamentally changing how computing works and what computers can do. The next industrial revolution has begun.

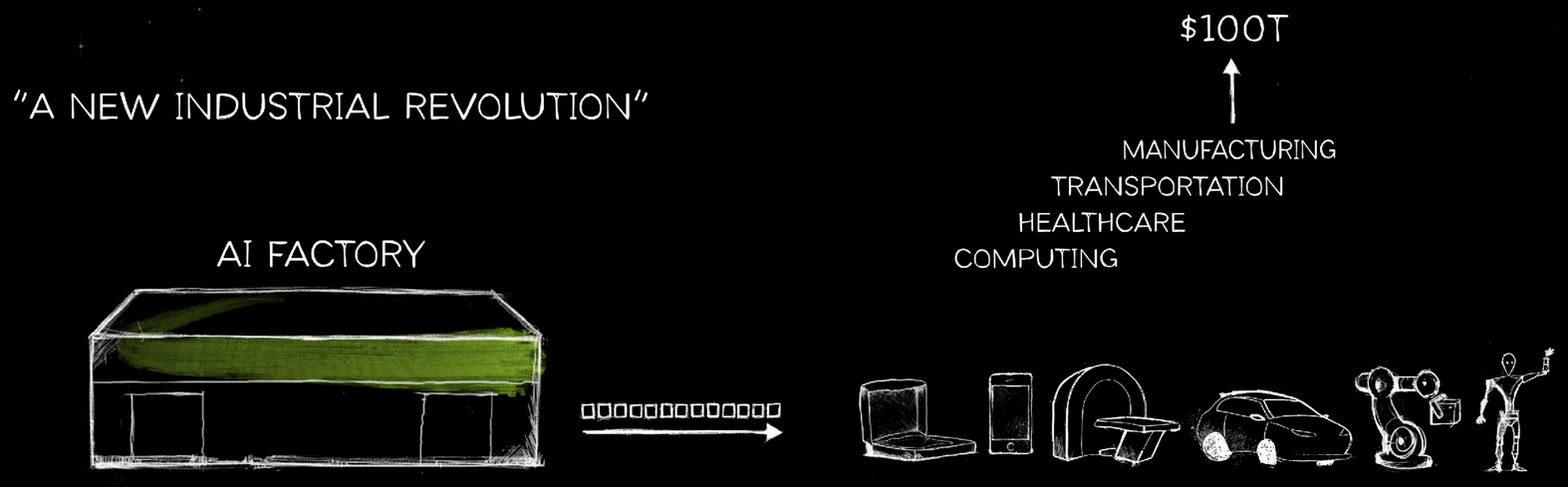
One of the most amazing things that anyone has ever said to me came from a quantum chemist -‘because of NVIDIA’s work, I can do my life’s work *in my lifetime*’

NVIDIA founder and CEO, Jensen Huang

Pioneering Accelerated Computing

Modern computing started sixty years ago with the IBM System/360. For the last decade—as the performance scaling of general-purpose processing has slowed while computing demand has continued—an exponentially growing performance-to-demand deficit has built up. NVIDIA accelerated computing has created a path forward at just the right time.

Accelerated computing starts with the most advanced processors and ends with AI factories. From chip architecture to advanced networking to acceleration libraries, NVIDIA builds the entire computing system at data center scale to produce intelligence at scale.



A New Computing Platform Sparks a New Industrial Revolution

Accelerated computing paved the way for generative AI. For the very first time, we can manufacture our most valuable resource—intelligence.

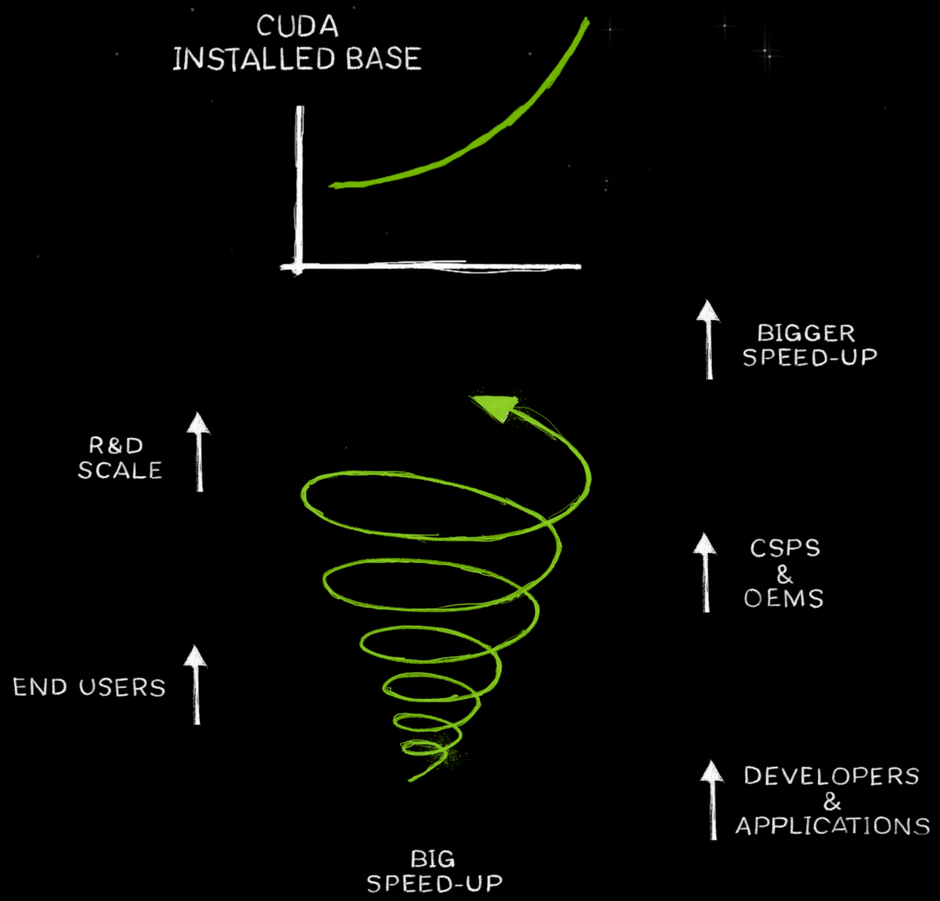
To seize this generational opportunity, companies and countries around the world are building NVIDIA-powered AI factories to process, refine, and manufacture intelligence from data. Intelligence, in the form of digital tokens, is a new kind of commodity, a new kind of product, and it will create new revenue opportunities for the world's $100 trillion of industries.

NVIDIA CUDA's Virtuous Cycle Propels AI Innovation

NVIDIA accelerated computing has reached a tipping point and achieved a virtuous cycle. The significant CUDA® installed base attracts developers and applications, which attracts resellers reaching customers, which expands the installed base to attract more developers.

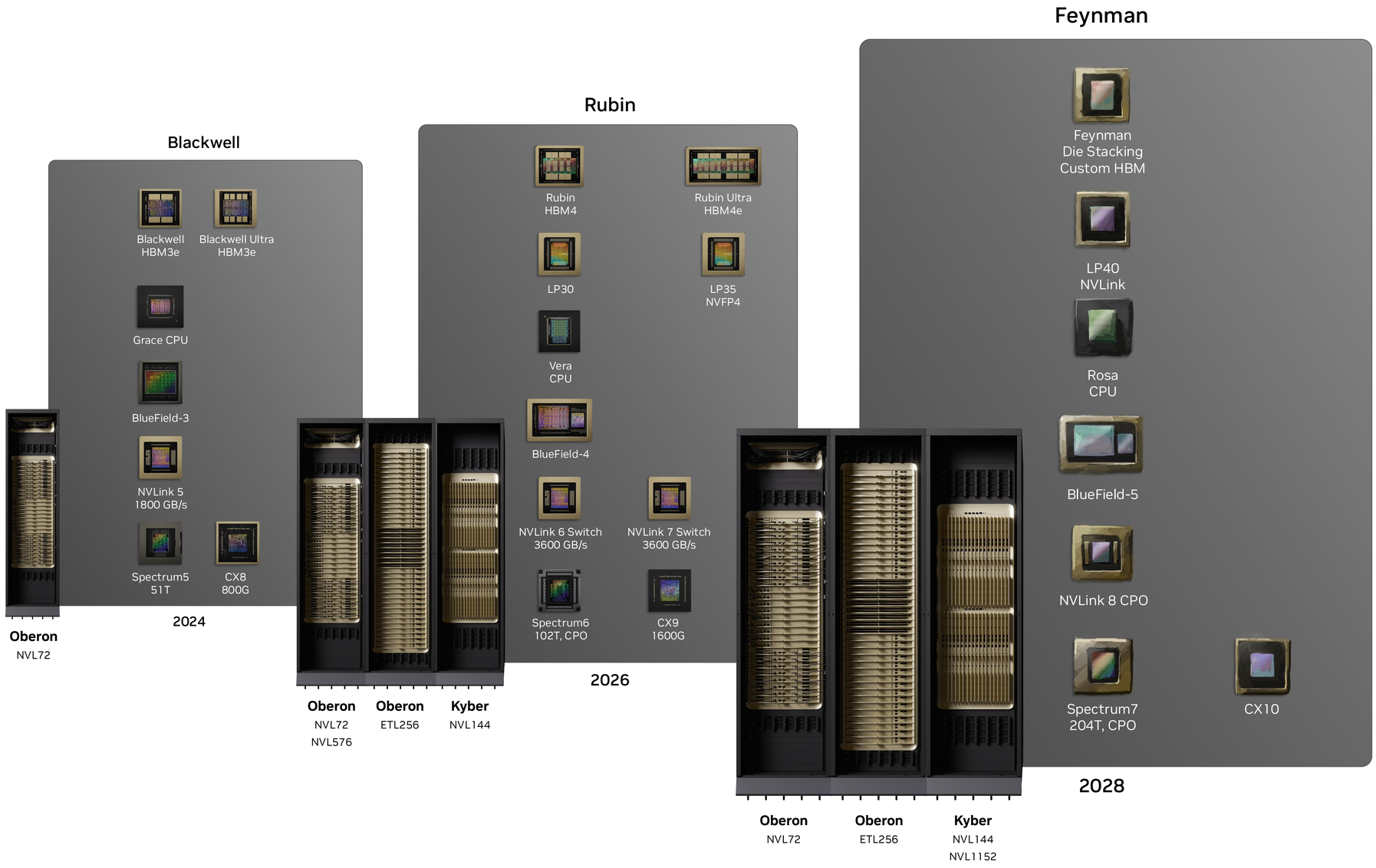
CUDA, our parallel computing model launched in 2006, offers developers an unparalleled toolkit with over 300 libraries, 600 AI models, numerous SDKs, and support for 3,700 GPU-accelerated applications. It has more than 53 million downloads.

The success of the CUDA model has led to the creation of a thriving ecosystem that now includes over 5 million developers, 40,000 companies, and thousands of generative AI companies—all building on the NVIDIA platform.



NVIDIA

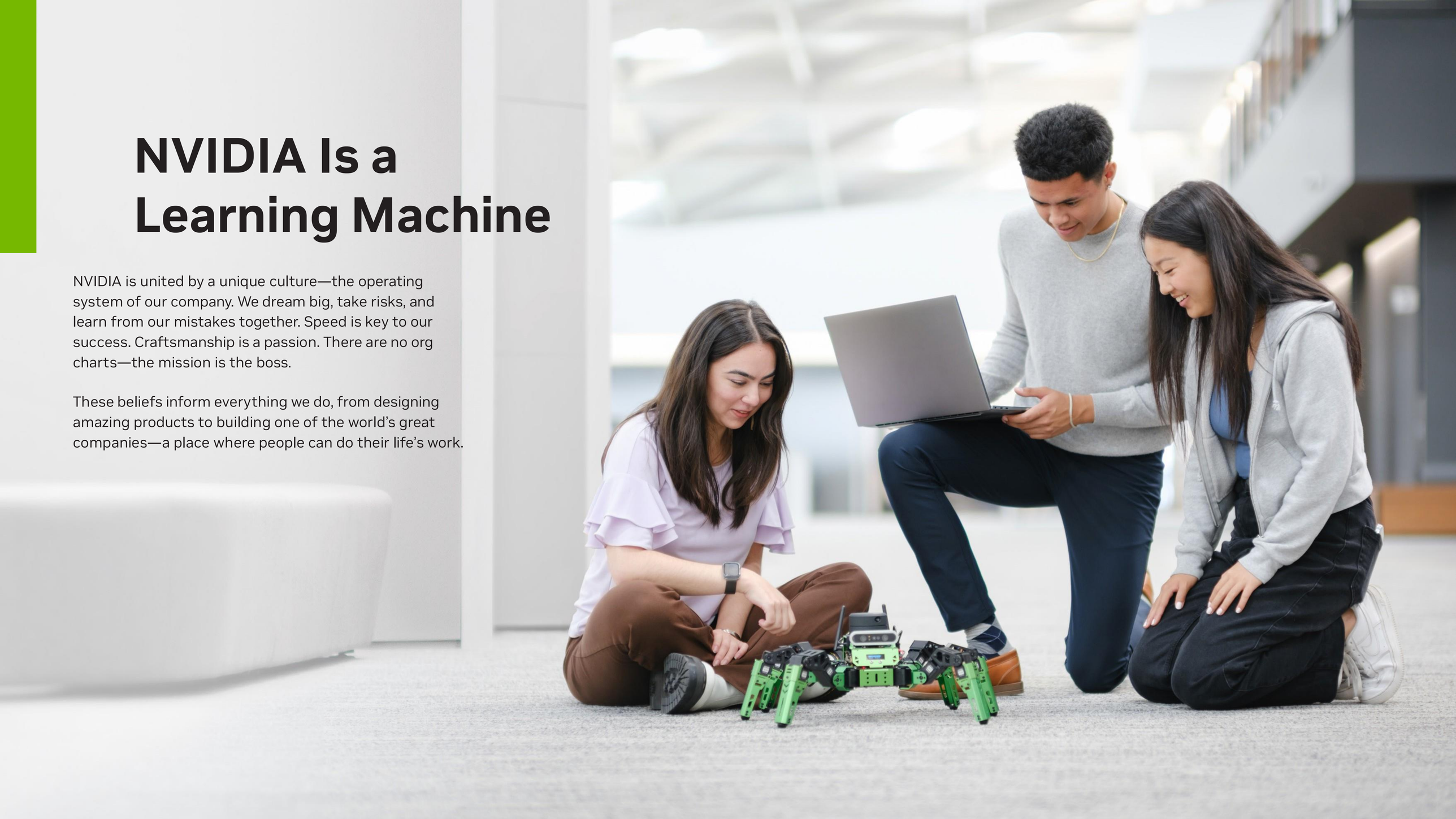
NVIDIA Extreme Co-Design Delivering X-Factors Every Year

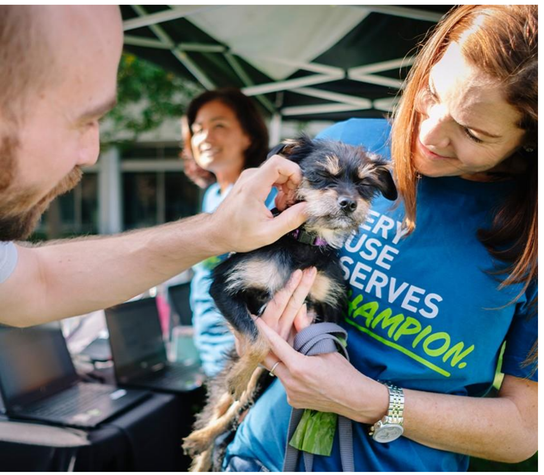


NVIDIA Is a Learning Machine

NVIDIA is united by a unique culture—the operating system of our company. We dream big, take risks, and learn from our mistakes together. Speed is key to our success. Craftsmanship is a passion. There are no org charts—the mission is the boss.

These beliefs inform everything we do, from designing amazing products to building one of the world's great companies—a place where people can do their life's work.



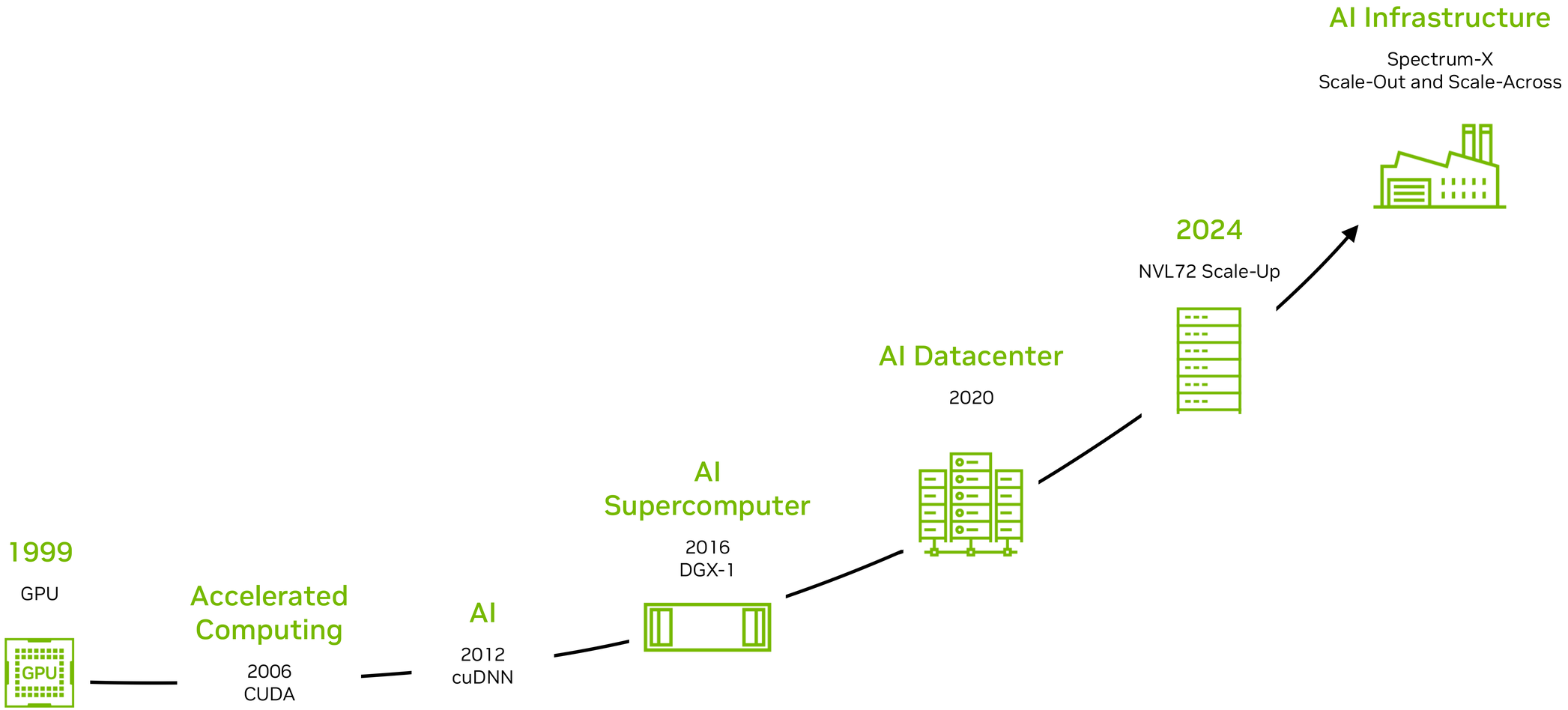


We’re One Team Tackling Challenges No One Else Can Solve

NVIDIA employees are dedicated to building technology that moves humanity forward and to supporting the communities in which they work and live.

We’ve been recognized as a top company in social responsibility, and our employees are passionate donors to hundreds of charities around the globe.

NVIDIA’s Evolution From Chips to an AI Infrastructure Company



20 Years of CUDA: The Foundation of Modern Science

Every Industry | One Architecture | CUDA Everywhere

7M
Developers

600M
Downloads

6K
Accelerated Applications

1K
SDKs and Models

cuLitho
Computational Lithography

cuDSS
cuSPARSE
cuFFT
AmgX
Computer-Aided Engineering

cuOpt
Decision Optimization

cuDF
cuVS
cuML
cuGraph
Data Science and Processing

MONAI
Medical Imaging

Megatron
Dynamo
NIXL
cuDNN
CUTLASS
Deep Learning

cuQuantum
CUDA-Q
Quantum Computing

cuEST
cuEquivariance
cuTensor
ALCHEMI
Quantum Chemistry

Aerial
Sionna
Holoscan
5G/6G
DAQUIRI
Signal Processing

Warp
Physics

Parabricks
Genomics

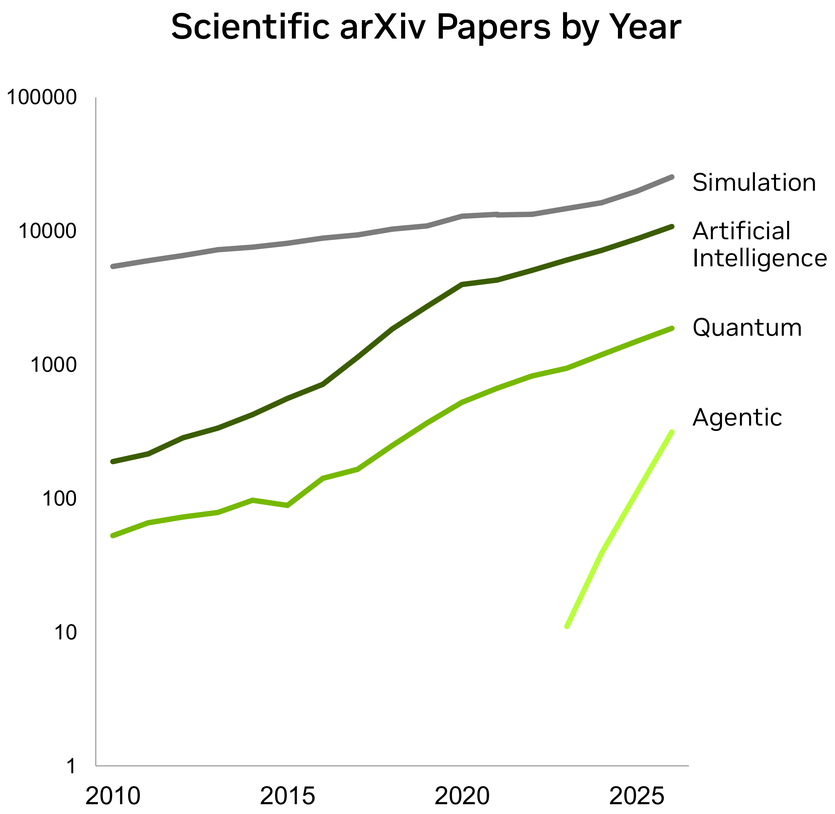
Earth-2
Weather Analytics

cuPyNumeric
cuPhoton
Numerical Computing

NVIDIA

Scientific Computing at an Inflection Point

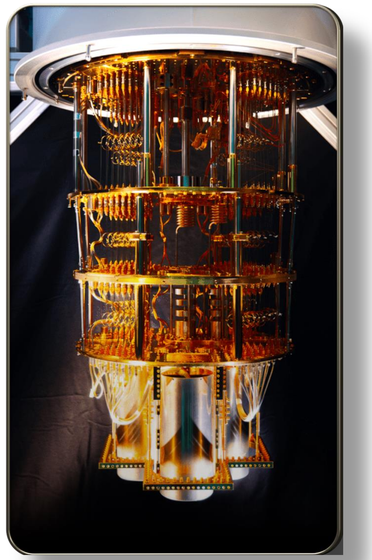
From One to Four Pillars of Scientific Computing



Simulation

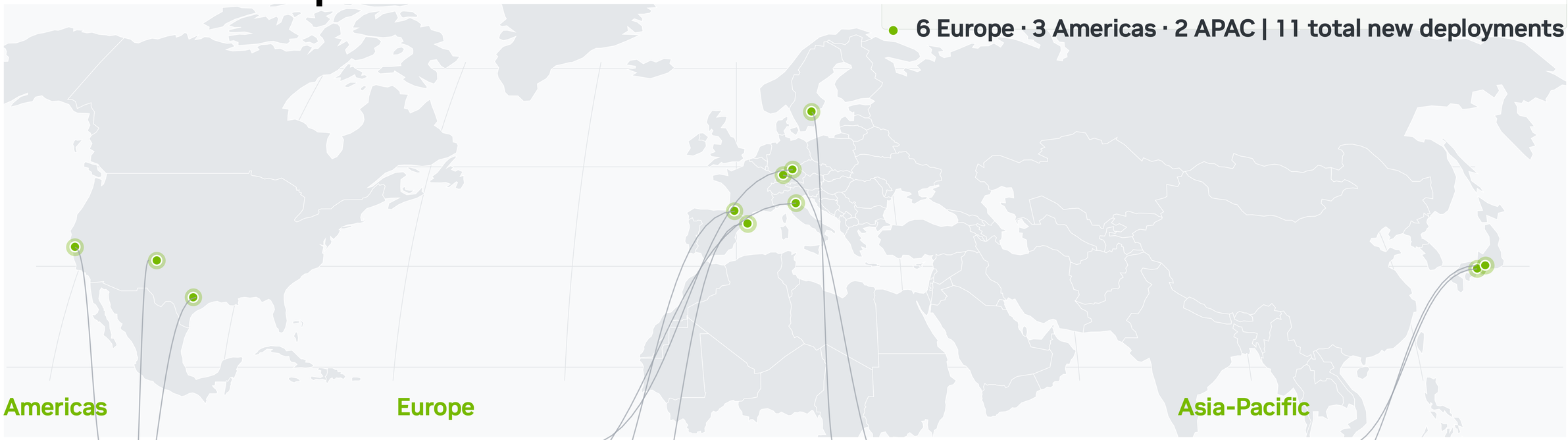
AI

Agentic



Quantum

ISC 2026: AI Factories and HPC Systems Coming Online Across Europe and the World



Americas

LANL ATS-5
USA | 2,160 VR NVL4 + 2,600 Vera CPUs

NERSC-10 Doudna
USA | Vera Rubin

TACC Horizon
USA | 4,000 GB200 NVL72 + 9,500 Vera CPUs

Europe

Bavarian AI / FAU Erlangen
Germany | 1,104 GB200 NVL4

CINECA IT4LIA
Italy | 8,000 GB200 NVL4 + 192 HGX-B200

TotalEnergies Pangea
France | 3,280 GB200 NVL4

BSC MareNostrum 5 AI Factory
Spain | 1,320 GB200 NVL4 + 504 GB300 NVL72

MIMEr
Sweden | 400 GB200 NVL4

HLRS HammerHAI
Germany | 860 GB200 NVL4

Asia-Pacific

RIKEN AI for Science
Japan | 1,000 GB200 NVL4

Nagoya University
Japan | 208 GB200 NVL4

OEM PARTNERS

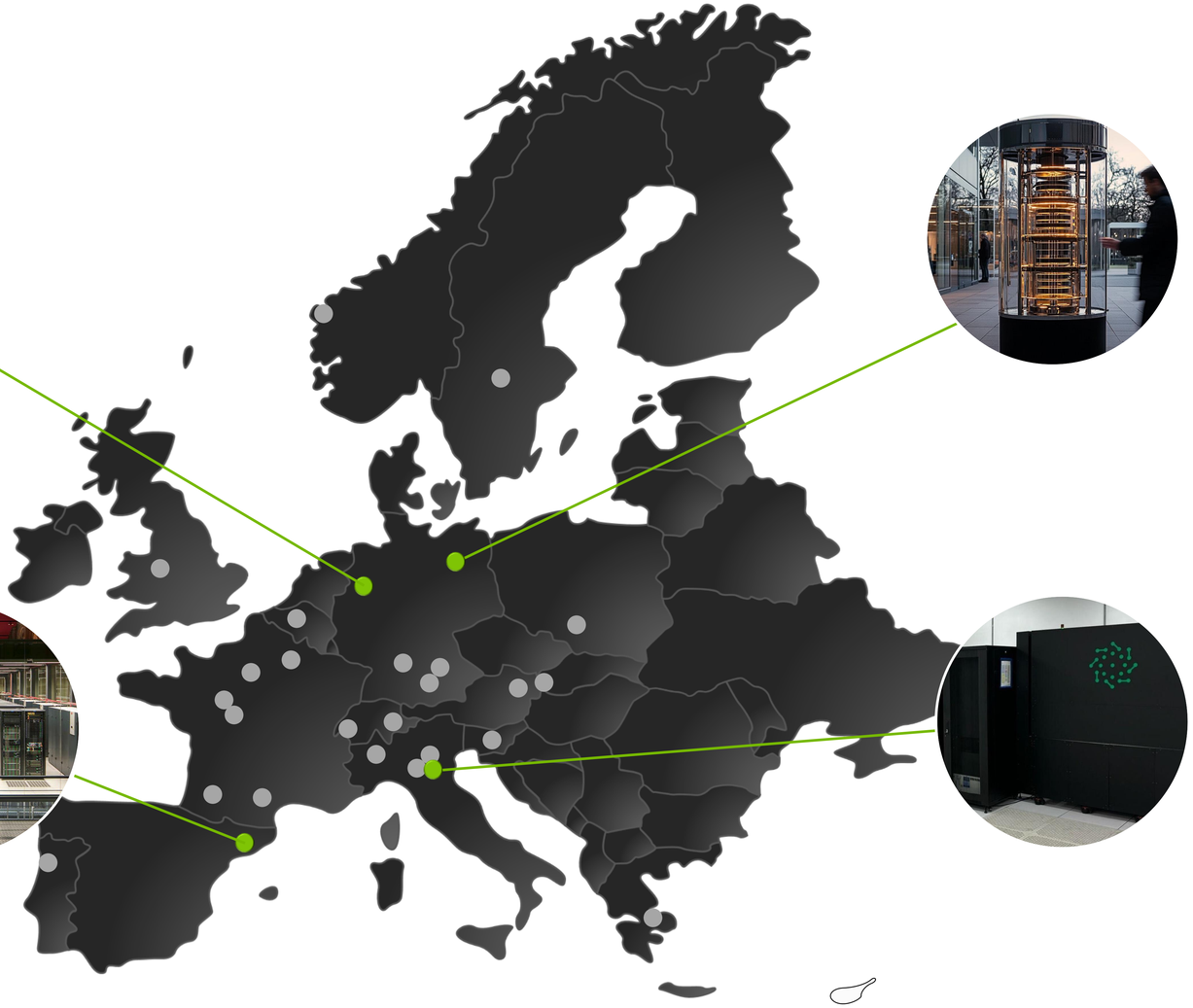
Bull · DELL Technologies · FUJITSU · GIGABYTE · HPE · Supermicro

NVIDIA

Europe Building Quantum-GPU Supercomputing on NVIDIA

JÜLICH FORSCHUNGSZENTRUM · D-Wave · IQM · Pasqal

Julich runs world-record 50 qubit quantum computing simulation



Fraunhofer FOKUS

Qrisp + CUDA-Q accelerates EU quantum applications

Barcelona Supercomputing Center
Centro Nacional de Supercomputación

QILIMANJARO QUANTUM TECH

BSC and Qilimanjaro announce Quantum-GPU integration with CUDA-Q

CINECA · Pasqal

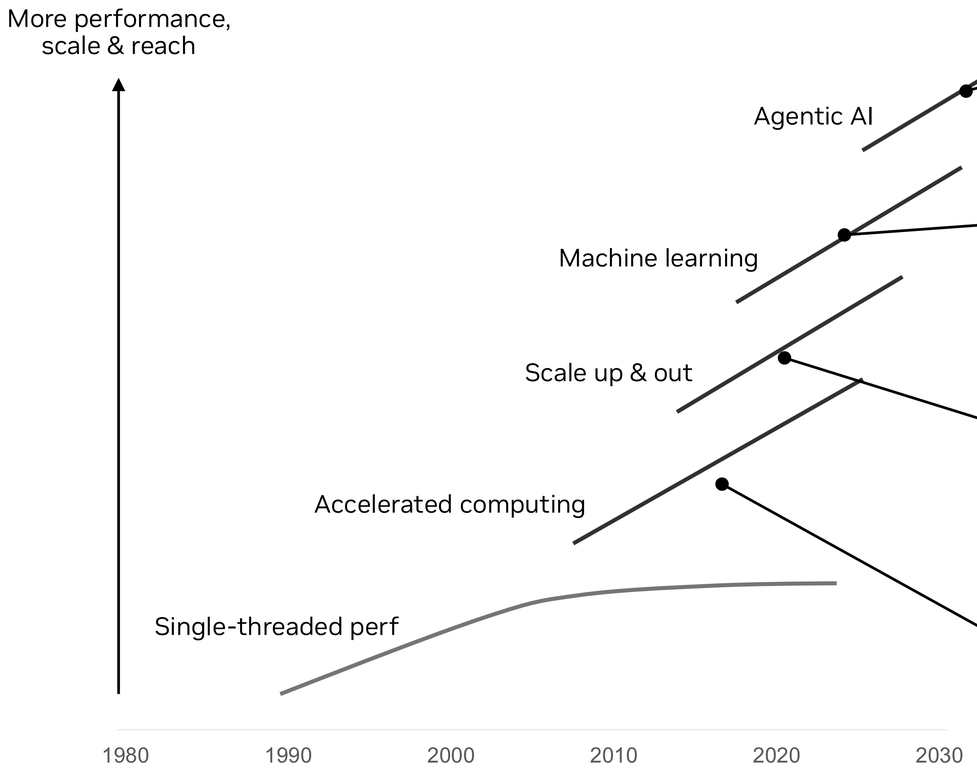
CINECA-Pasqal quantum integration inaugurated with CUDA-Q

NVIDIA

AI for Science

Incredible Growth in Scale and Reach of Scientific Computing

Expanding capabilities, accessibility and applications leading to new discoveries



Increasing capabilities and access for scientific computing

OpenFold: Retraining AlphaFold2 yields new insights into its learning mechanisms and capacity for generalization

Gustaf Ahdritz[*1,2], Nazim Bouatta[*3], Christina Floristean[1], Sachin Kadyan[1], Qinghui Xia[1], William Gerecke[3], Timothy J O'Donnell[4], Daniel Berenberg[5,6], Ian Fisk[7], Niccolò Zanichelli[8], Bo Zhang[9], Arkadiusz Nowaczynski[10], Bei Wang[10], Marta M Stepniewska-Dziubinska[10], Shang Zhang[10], Adegoke Ojewole[10], Murat Efe Guney[10], Stella Biderman[11,12], Andrew M Watkins[5], Stephen Ra[5], Pablo Ribalta Lorenzo[10], Lucas Nivon[13], Brian Weitzner[14], Yih-En Andrew Ban[15], Peter K Sorger[3], Emad Mostaque[16], Zhao Zhang[17],

Universal Quantum Computer Simulation of 50 Qubits on Europe's First Exascale Supercomputer Harnessing Its Heterogeneous CPU-GPU Architecture

Hans De Raedt[a], Jiri Kraus[b], Andreas Herten[a], Vrinda Mehta[a,*], Mathis Bode[a], Markus Hrywniak[b], Kristel Michielsen[a], Thomas Lippert[a]

[a] *Jülich Supercomputing Centre, Forschungszentrum Jülich, D-52425 Jülich, Germany*
[b] *NVIDIA, Würselen, Germany*

Abstract

We have developed a new version of the high-performance Jülich universal quantum computer simulator (JUQCS-50) that leverages key features of the GH200 superchips as used in the JUPITER supercomputer, enabling simulations of a 50-qubit universal quantum computer for the first time. JUQCS-50 achieves this through three key innovations: (1)

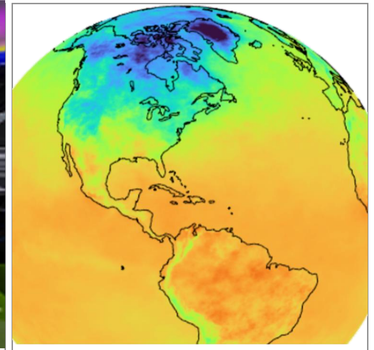
Breaking the Million-Electron and 1 EFLOP/s Barriers: Biomolecular-Scale *Ab Initio* Molecular Dynamics Using MP2 Potentials

Ryan Stocks, *School of Computing, Australian National University*, ryan.stocks@anu.edu.au
Jorge L. Galvez Vallejo, *School of Computing, Australian National University*, jorge.galvezvallejo@anu.edu.au
Fiona C. Y. Yu, *School of Computing, Australian National University*, fiona.yu@anu.edu.au
Calum Snowdon, *School of Computing, Australian National University*, calum.snowdon@anu.edu.au
Elise Palethorpe, *School of Computing, Australian National University*
Jakub Kurzak, *Advanced Micro Devices, Inc.*, jakub.kurzak@amd.com
Dmytro Bykov, *Computing for Chemistry, Oak Ridge National Lab*, bykovd@ornl.gov
Giuseppe M. J. Barca, *School of Computing, University of Melbourne*

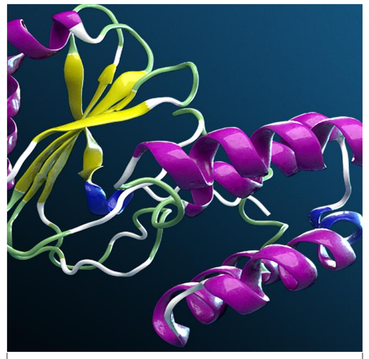
NVIDIA AI Accelerates Scientific Discovery

With Traditional Science, Discovery Takes Years

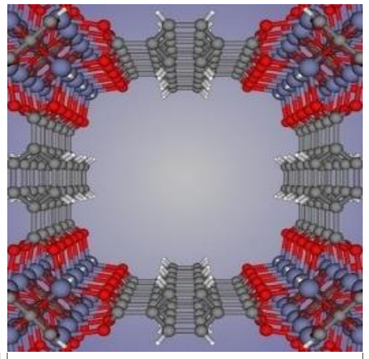
AERODYNAMICS



CLIMATE AND WEATHER



DRUG DISCOVERY



MATERIALS DISCOVERY

Across Domains, AI Shrinks Time-to-Science

Chemistry
Demis Hassabis, John Jumper for AlphaFold
David Baker for Rosetta (protein design)

Physics
Geoff Hinton and John Hopfield
for artificial neural networks

Groundbreaking Achievements with NVIDIA

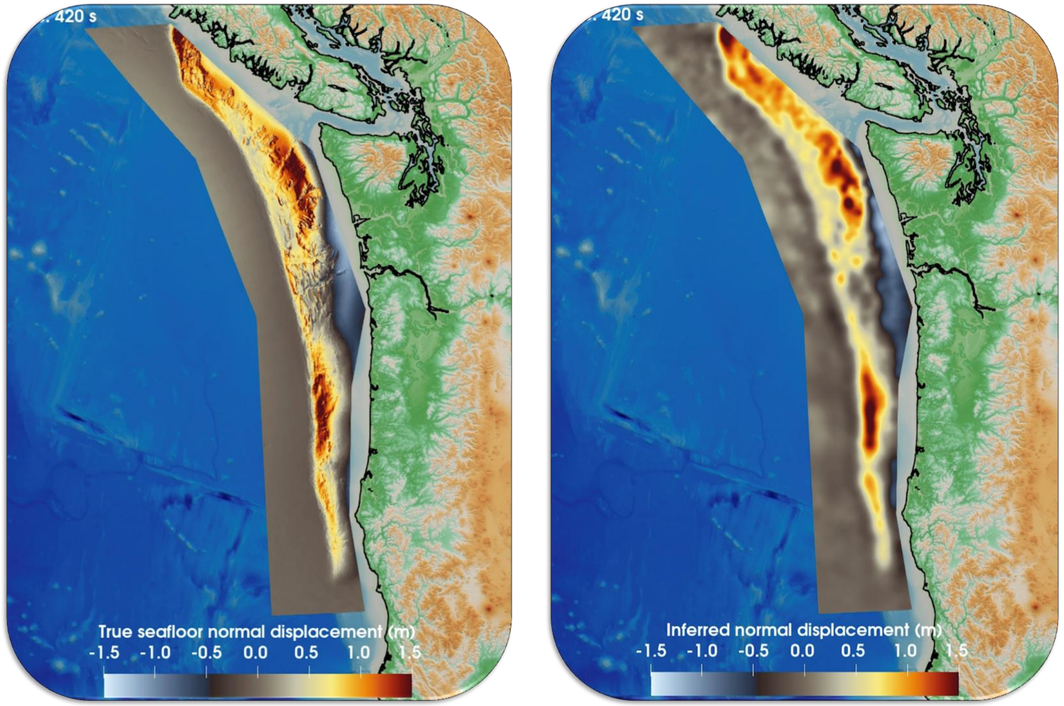
Photo by Hans Reiners on Unsplash

NVIDIA

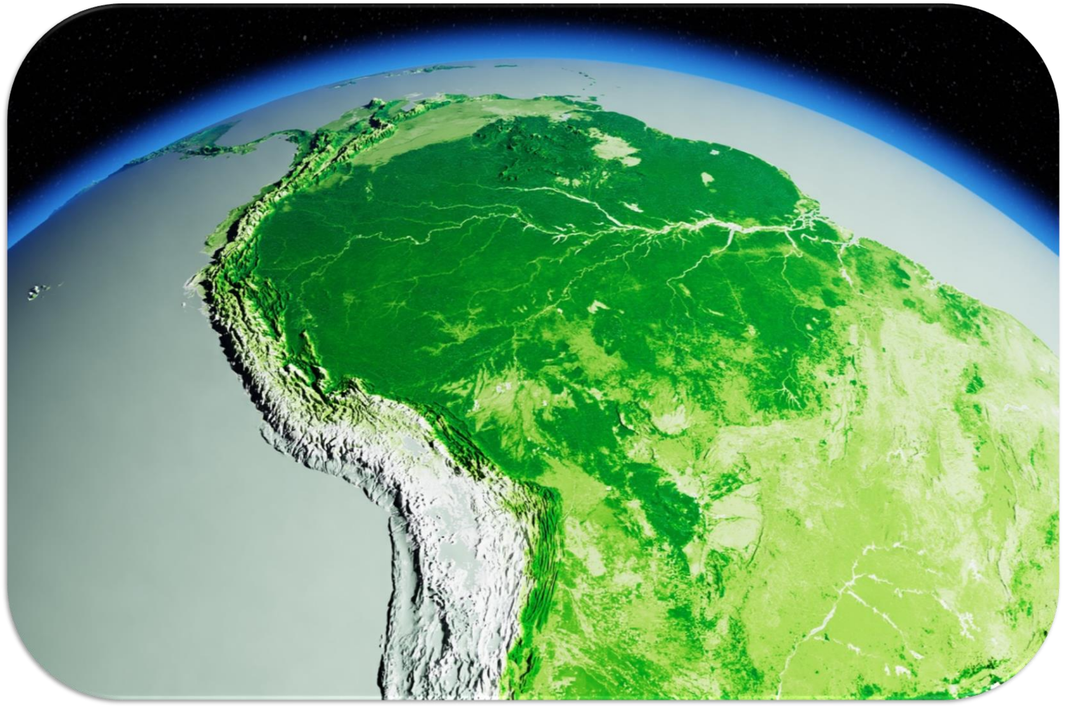
2025 Gordon Bell Winners

Gordon Bell winners pushed science boundaries

DIGITAL TWIN FOR TSUNAMI FORECAST
Omar Ghattas (University of Texas Austin) et al.



FULL EARTH SYSTEM AT 1KM
Daniel Klocke (MPI-M) et al.

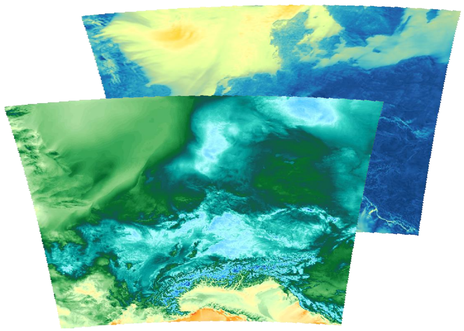


NVIDIA Earth-2 Platform

Accelerated weather and climate modeling and with high-resolution interactive visualization

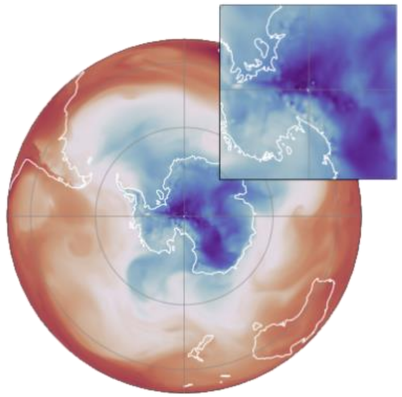
NWP Simulation

Accelerated numerical simulation of weather and climate physics.



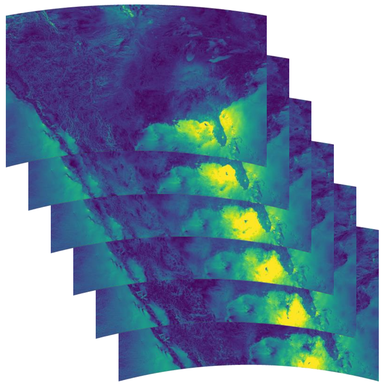
AI Training

Training of AI models for forecasting, downscaling, interpolation, and other applications.



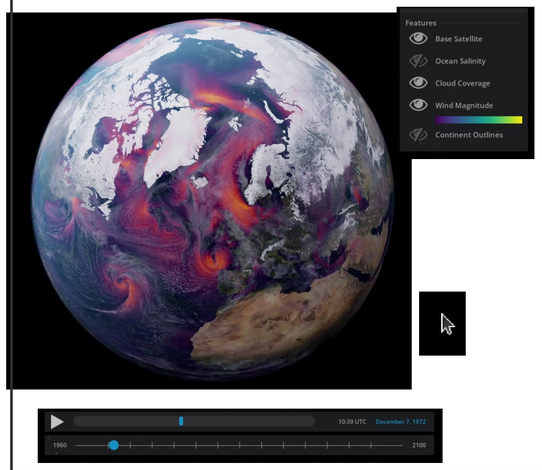
AI Inference

Efficient inference, connecting AI models, data sources, and downstream applications.



Visualization

Interactive, visualization and scalable with photorealistic rendering.



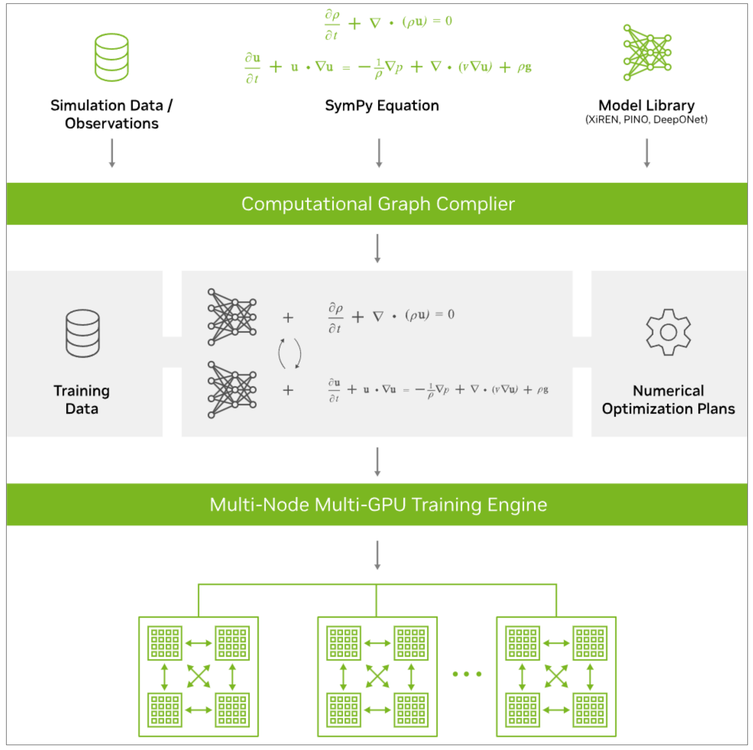
Data Federation

Earth-2: AI Training & Inference Enhancement

Best technology to accelerated DLWPs

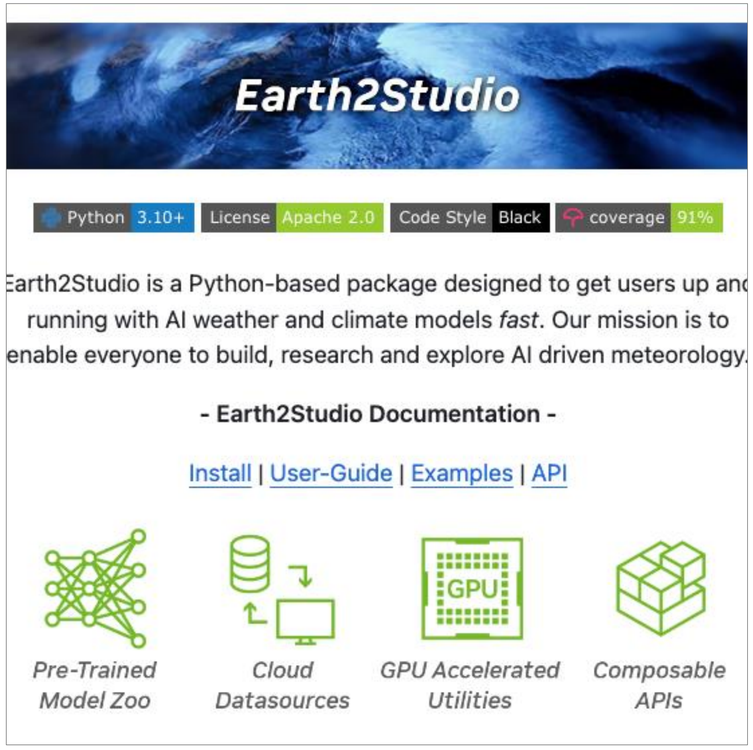
Makani – Research with Massive Parallel Training

https://github.com/NVIDIA/makani



PhysicsNeMo - Training

https://github.com/NVIDIA/physicsnemo

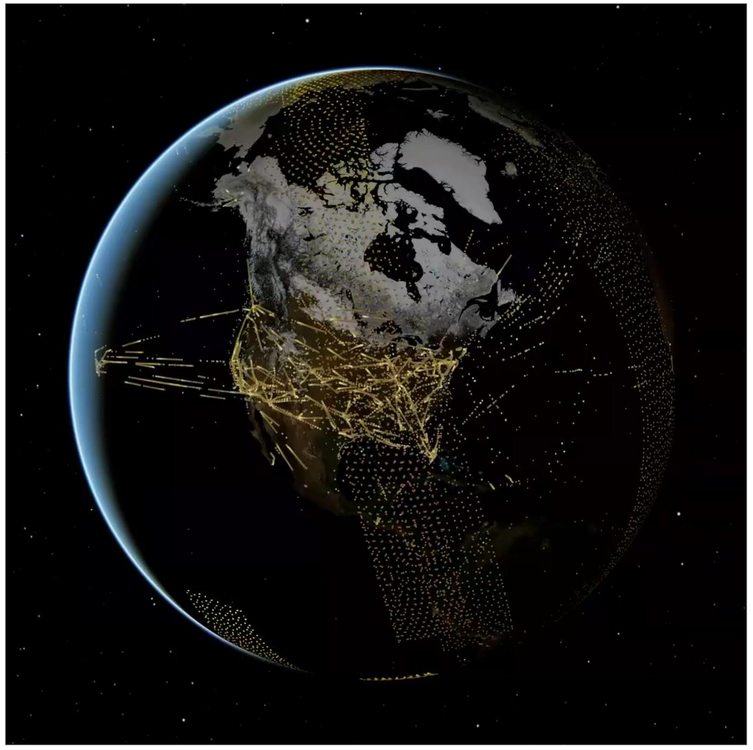


Earth2Studio - Exploration

https://github.com/NVIDIA/earth2studio

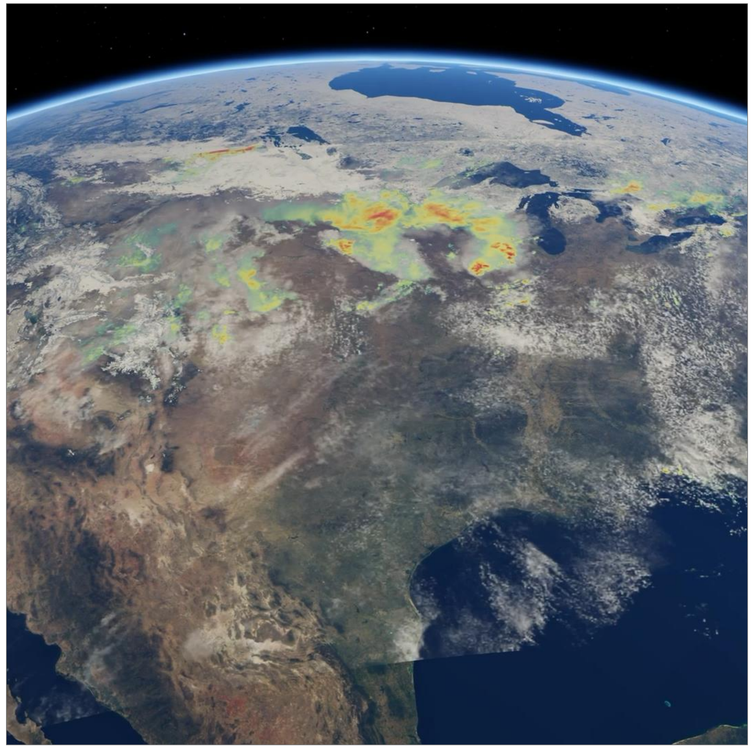
AMS 2026 Earth-2 Family of Open Models

https://huggingface.co/blog/nvidia/earth-2-open-models



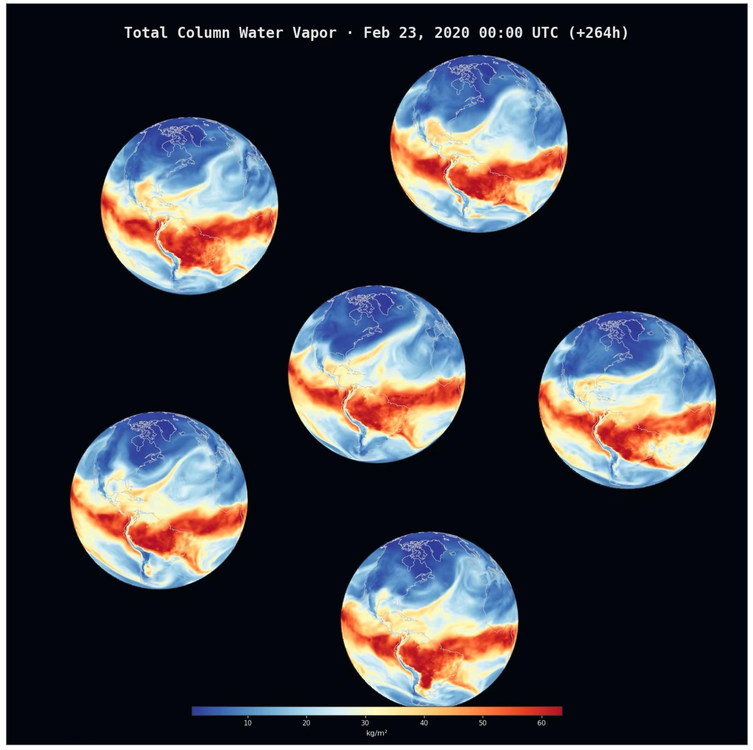
Earth-2 Global DA - HealDA

https://research.nvidia.com/publication/2026-01_healda-highlighting-importance-initial-errors-end-end-ai-weather-forecasts



Earth-2 Nowcasting - StormScope

https://research.nvidia.com/publication/2026-01_learning-accurate-storm-scale-evolution-observations



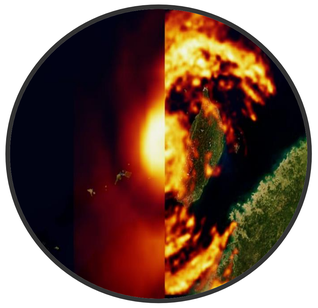
Earth-2 Medium Range – Atlas

https://research.nvidia.com/publication/2026-01_demystifying-data-driven-probabilistic-medium-range-weather-forecasting

NVIDIA

Broad Portfolio of Open Weather Models

Unlocking breakthroughs with revolutionary approaches to AI weather forecasting

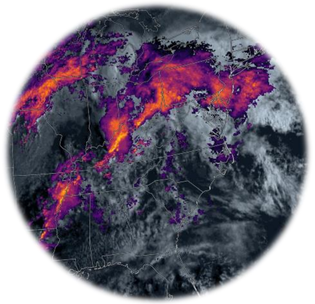


CorrDiff

Super resolution of course forecasting

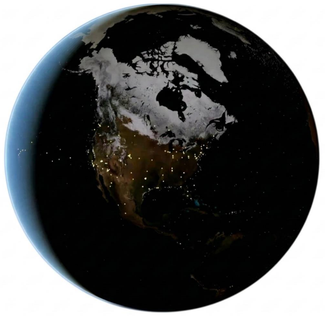
Earth-2 Medium Range

15-day forecasts with breakthrough accuracy



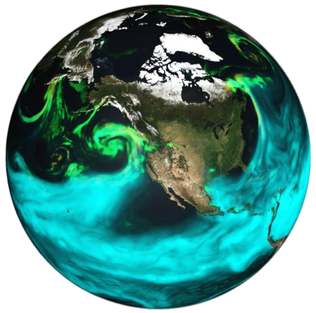
Earth-2 Nowcasting

Enables rapid forecast updates with physics-based model accuracy



Earth-2 Global Data Assimilation

Generate initial conditions from multi-source, multi-modal data



FourCastNet

Medium range, global ensemble forecasting

How Does AI Help Identify useful Chemicals and Materials?

Traditional Chemicals Discovery

Human Intuition

Traditional Computations

Testing required with Trial and Error

Years

AI-driven Chemicals Discovery

AI + Human Intuition

AI Computations

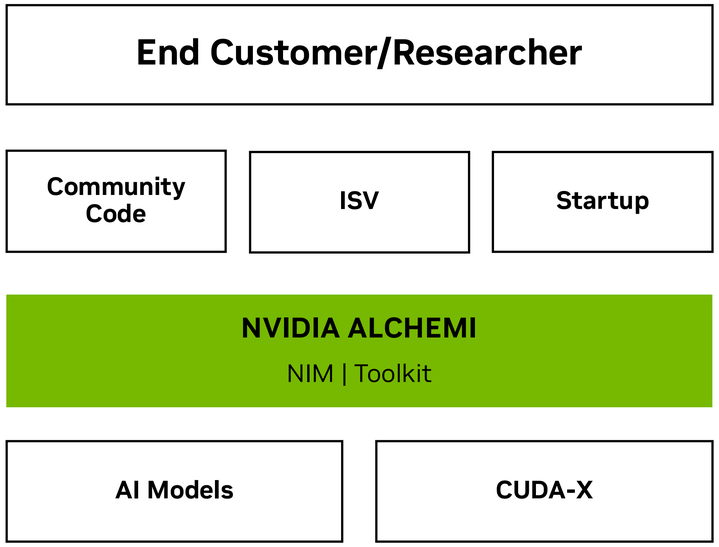
Testing required with AI-guided Search

Months

Accelerated Innovation powered by NVIDIA

NVIDIA ALCHEMI

Usability, flexibility and performance



ALCHEMI Toolkit: **50k+ downloads** per month since Dec 2025 launch

GitHub: https://github.com/NVIDIA/nvalchemi-toolkit-ops

What is it

NVIDIA ALCHEMI (AI Lab for Chemistry and Materials Innovation) is a collection of domain-specific NIM microservices and Toolkit for accelerating chemical and materials discovery (e.g., battery, catalyst, OLED, beauty product):

- **NIM**: Scalable layer of cloud-ready microservices for chemistry and materials science, enabling deployment and orchestration on NVIDIA-accelerated platform
- **Toolkit**: Collection of GPU-accelerated batched kernels and tools for training Machine Learning Interatomic Potentials (MLIPs) and running accelerated batched atomistic simulations with MLIPs

Who is it for

- **Researchers**: Computational chemists and materials scientists in Industry and Academia requiring atomistic simulation at near quantum chemistry accuracy to inform chemical and materials discovery
- **Developers**: Community code, ISV[1] and AI-first startup developers training MLIPs and building accelerated batched MLIP atomistic simulation engines

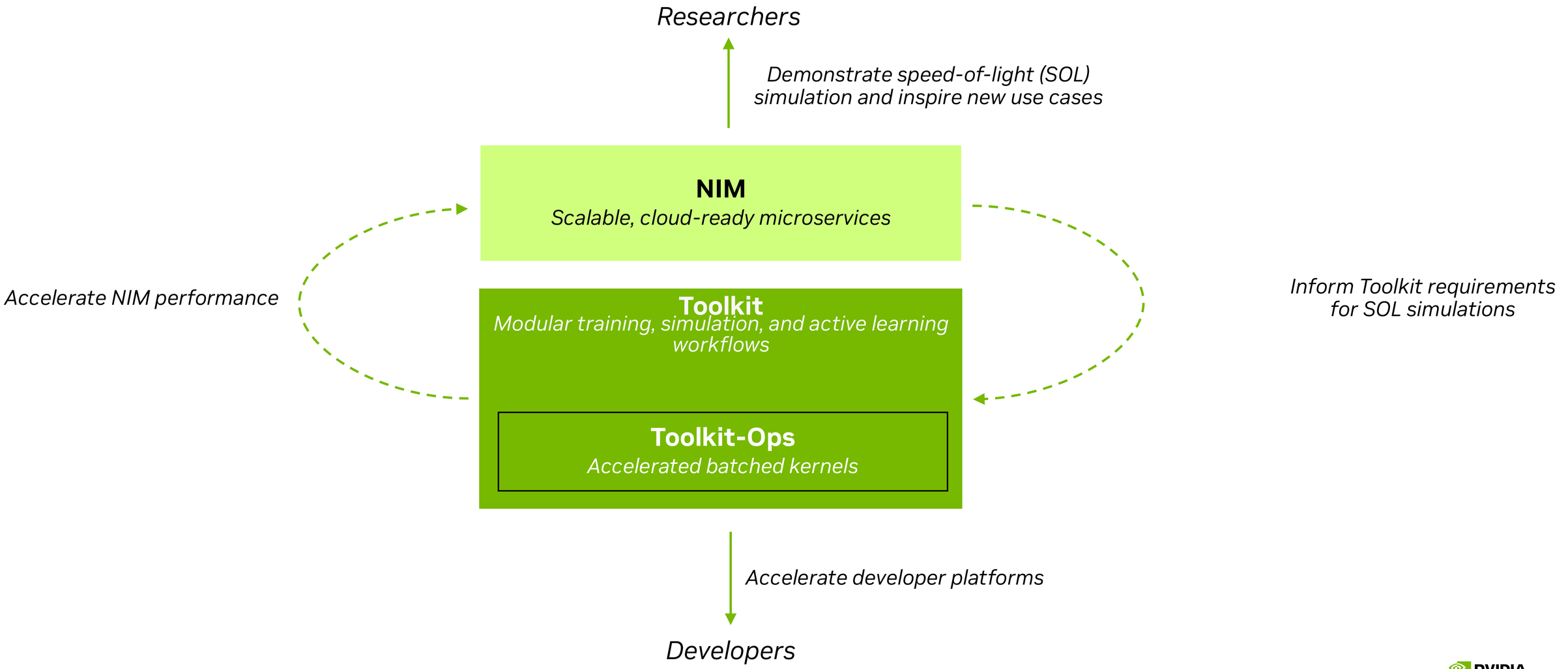
Value proposition

Accelerate chemical and materials discovery by 10X by enabling accurate, high performance atomistic simulations with AI and increasing successful outcome in lab

1. Independent software vendor

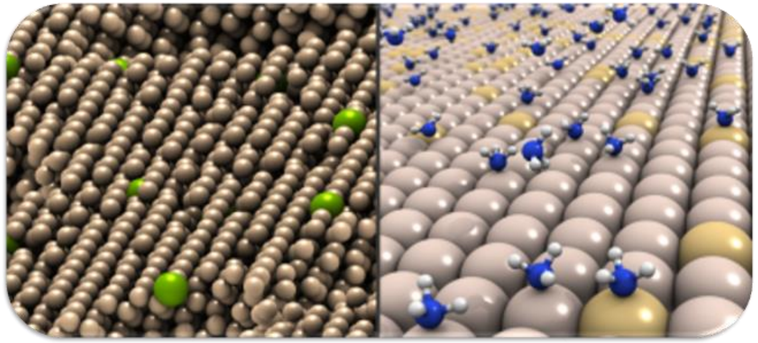
Tiered, Synergistic Offerings: Composable Workflows, Fast Primitives

Entry points for users at every level



Industry Examples

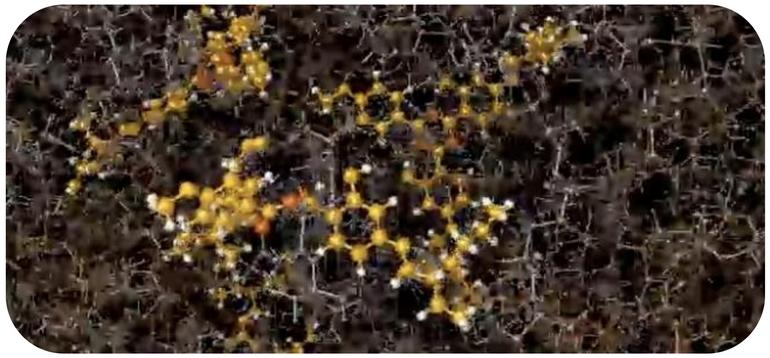
ALCHEMI enables end use cases across a diverse range of industries and applications



Catalysts for Hydrogen Production
10x acceleration of discovery timelines

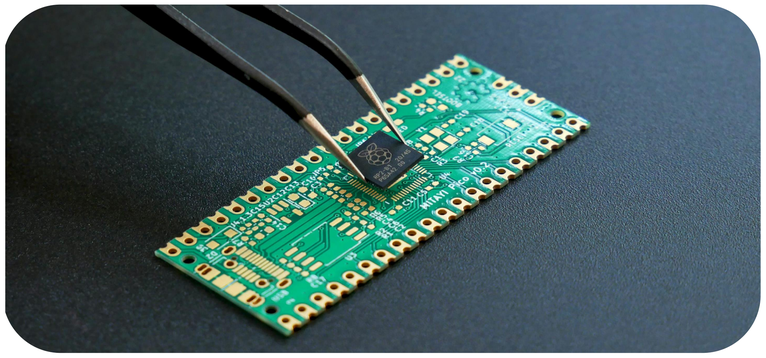
Molecules for OLEDs
10,000x acceleration vs. simulation methods

Electrolyte Materials for Li Batteries
800x acceleration in MLIP calculations



Co-Solvents for Skincare Formulations
100x faster screening for sun protection

Non-Rare-Earth Permanent Magnets
Up to 50x acceleration in magnetic materials screening



Deposition in semiconductor manufacturing
10-20X acceleration observed over popular OSS MD package

NVIDIA BioNeMo Platform

Open developer platform for AI-driven life sciences

Domain Specific

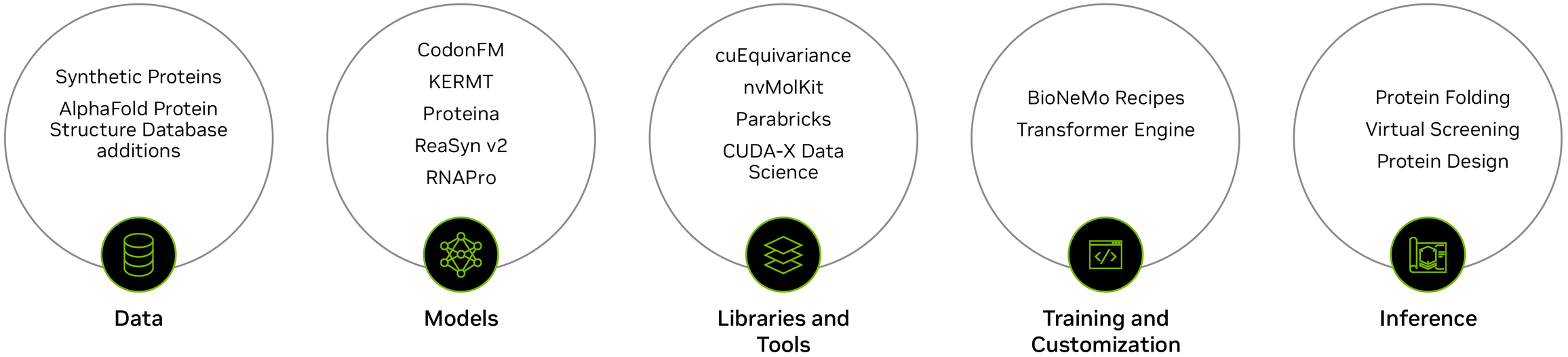
BioNeMo addresses the challenges in accelerating solutions in life sciences and drug discovery.

Open

Open by design, open-source and freely available. Develop and design for your biological intelligence.

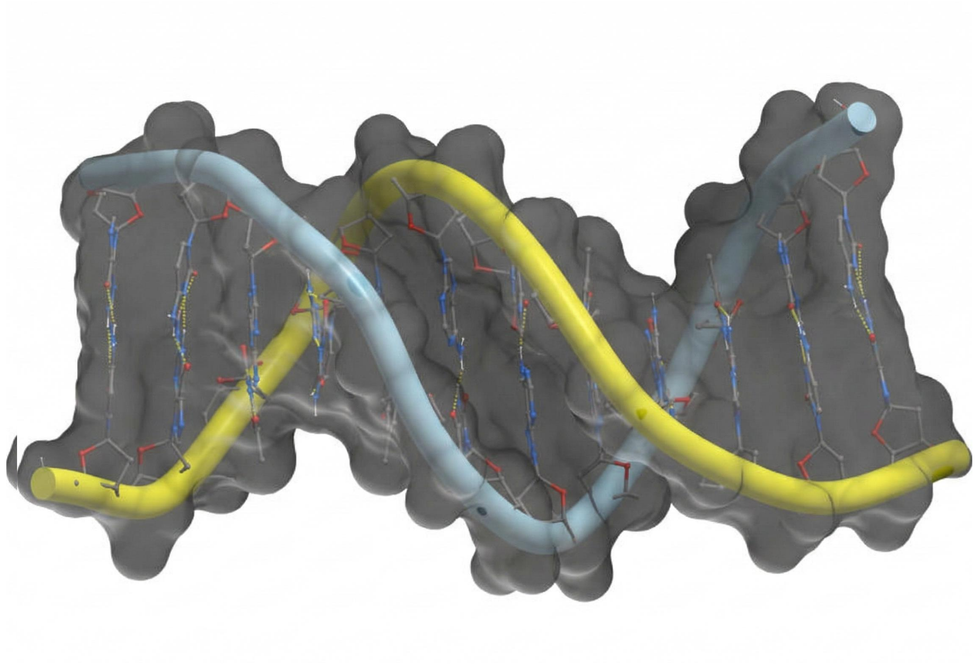
Accelerated

Co-designed with NVIDIA's full stack AI infrastructure, best TCO.



NVIDIA EVO 2 NIM Microservice

NVIDIA Evo 2 NIM microservice can generate a variety of biological sequences



```
Define JSON example human L1 retrotransposable element sequence
example = {

# nucleotide sequence to be analyzed
      "sequence":
"GAATAGGAACAGCTCCGGTCTACAGCTCCCAGCGTGAGCGACGCAGAAGACGGTGATTTCTGCATTTCCATCTGAGGTACCGGGTTCATCTCACTAGGGAG
TGCCAGACAGTGGGCGCAGGCCAGTGTGTGTGCGCACCGTGCGCGAGCCGAAGCAGGGCGAGGCATTGCCTCACCTGGGAAGCGCAAGGGGTCAGGGA
GTTCCCTTTCCGAGTCAAAGAAAGGGGTGATGGACGCACCTGGAAAATCGGGTCACTCCCACCCGAATATTGCGCTTTTCAGACCGGCTTAAGAAACGGCG
CACCACGAGACTATATCCCACACCTGGCTCAGAGGGTCCTACGCCCACGGAATC",
      "num_tokens": 102, # number of tokens to generate
      "top_k": 4, # only predict top 4 most likely outcomes per token
      "top_p": 1.0, # include 100% cumulative prob results in sampling
      "temperature": 0.7, # add variability (creativity) to predictions
      "
": True, # enable more diverse outputs
      "enable_logits": False, # disable raw model output (logits)
}

# Retrieve the API key from the environment
key = os.getenv("NVCF_RUN_KEY")

# Send the example sequence and parameters to the Evo 2 API
r = requests.post(

      # Example URL for the Evo 2 model API.
      url=os.getenv("URL","https://health.api.nvidia.com/v1/biology/arc/evo2-40b/generate")

      # Authorization headers to authenticate with the API
      headers={"Authorization": f"Bearer {key}"},

      # The data payload (sequence and parameters) sent as JSON
      json=example,
)
```

EVO-2: 90% predictive accuracy distinguishing benign and pathogenic mutations

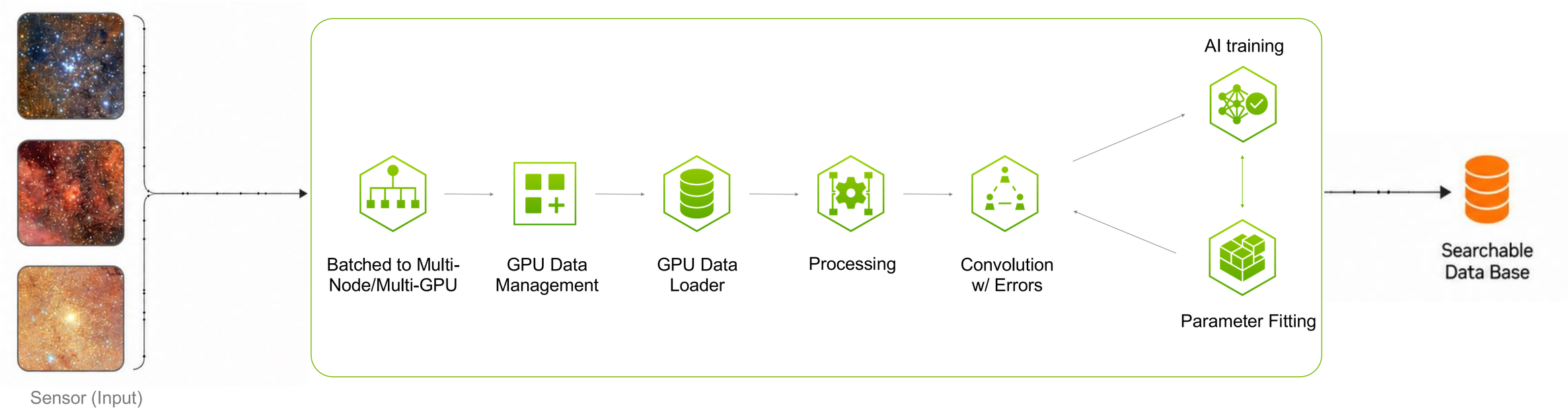
40B Model Throughput is up to 33 nucleotides per second using EVO-2 NIM

NVIDIA

NVIDIA cuPhoton

New library for telescopes, X-rays and laser experiments

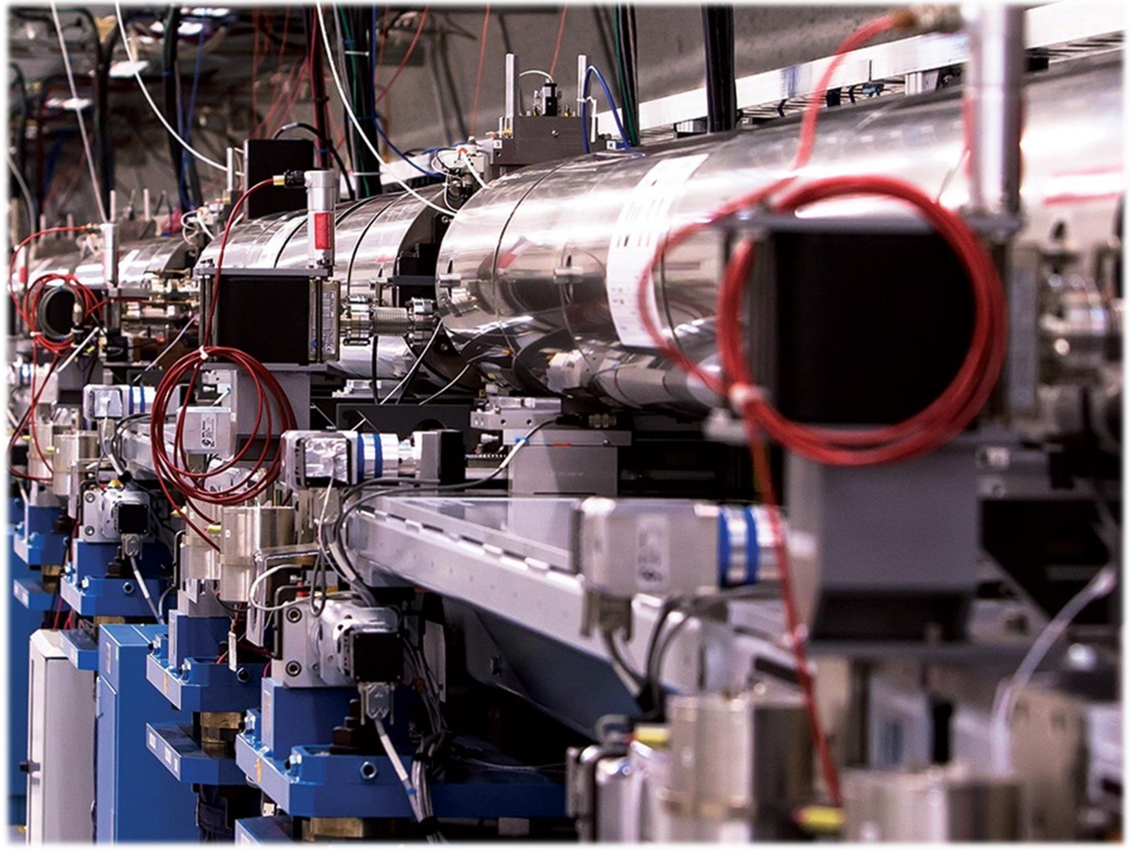
Enables analysis of full camera image in real time for scientific discoveries



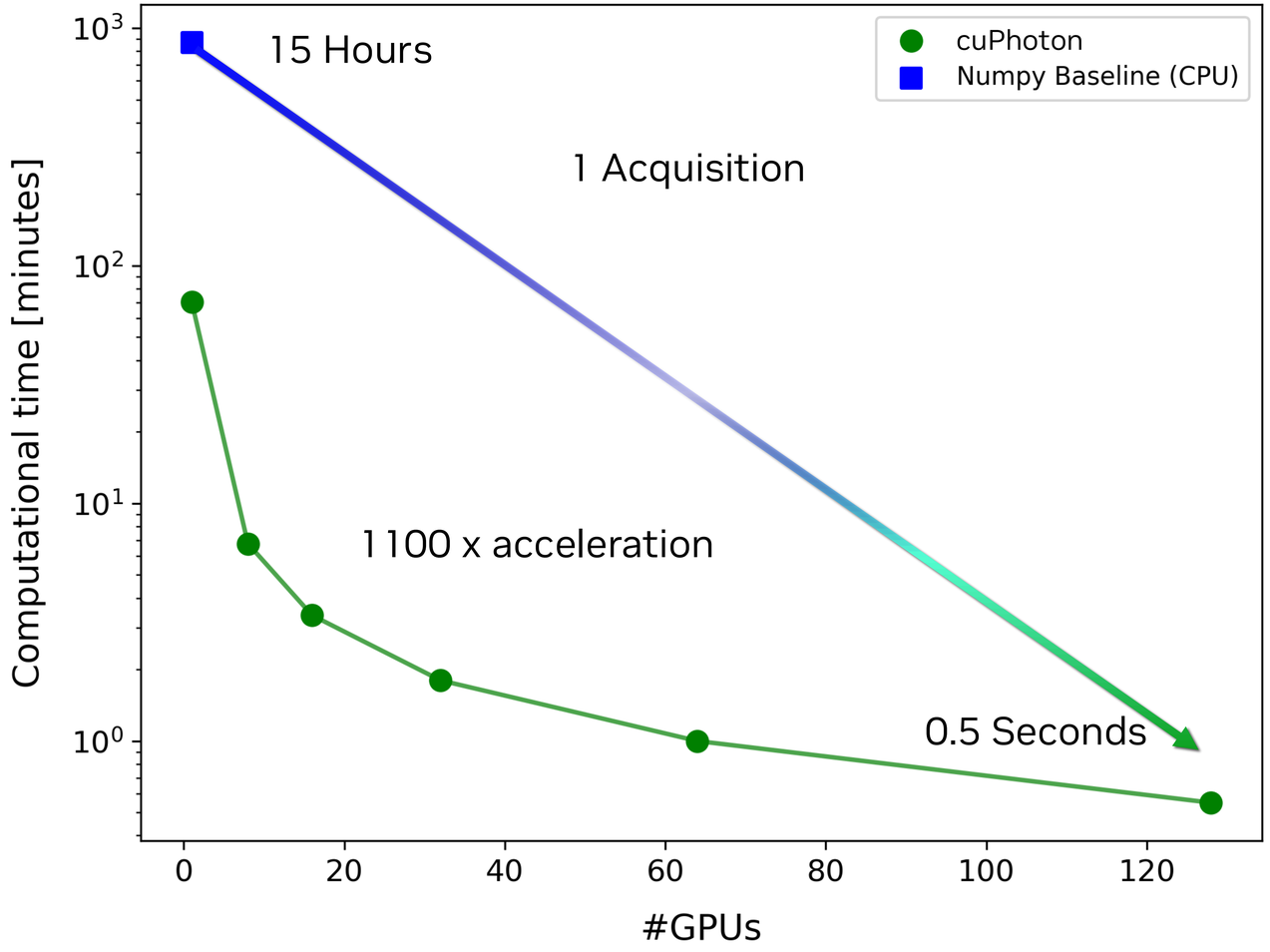
Accelerates image loading/reading by up to 15,000x, and processing and analysis by up to 8,000x

NVIDIA Accelerates X-ray Experiments over 1100 X

Analyzed 42 Terabytes multidimensional data in 4 hours instead of 9 months using cuPhoton, H100s



Linac Coherent Light Source



NVIDIA

Vera Rubin

Leading Supercomputing Centers and Global System Manufacturers Embrace Vera Rubin

Powering modelling, simulation, data science and AI across science domains

Bull

DELL

GIGABYTE™

Hewlett Packard Enterprise

SUPERMICRO

System Manufacturers

2,160 Rubin GPUs + 1,080 Vera CPUs

LANL

1,296 Rubin GPUs + 648 Vera CPUs

NEW at ISC26

576 Rubin GPUs + 288 Vera CPUs

NERSC

10x faster than Perlmutter

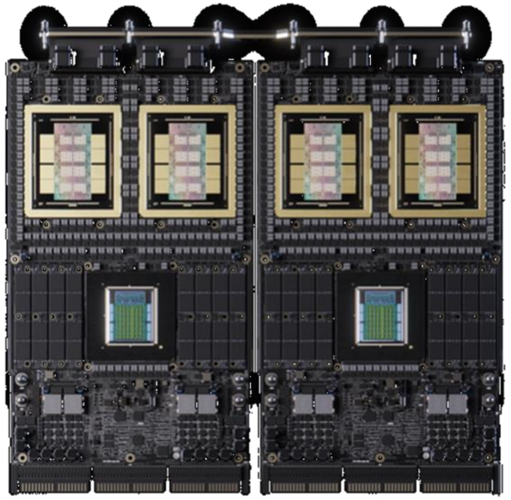
LRZ

30x faster than SuperMUC-NG

NVIDIA

NVIDIA Vera Rubin Delivers World-Class Supercomputers for Science

Up to 144 GPUs in a single rack for high-precision simulation and AI



	Vera Rubin Supercomputing Rack (144 Rubin GPUs)
Native FP64	5 PFLOPS
FP64 Matrix Multiply	30 PFLOPS
NVFP4 Inference	7.2 EFLOPS
NVFP4 Training	5 EFLOPS
HBM4 Bandwidth	3 PB/s
HBM4 Capacity	41 TB

All Specs are preliminary and subject to change.

NVIDIA

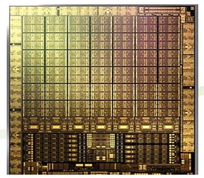
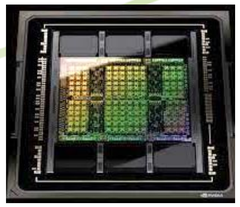
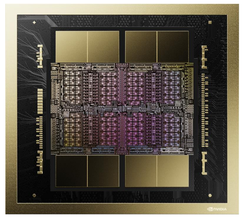
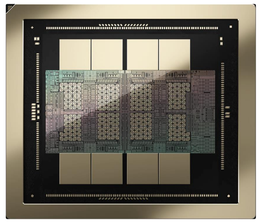
NVIDIA Fully Committed to Supporting Science

Precisions Needed for Scientific Simulation Including FP64 Are Supported

Scientific Computing is in our DNA

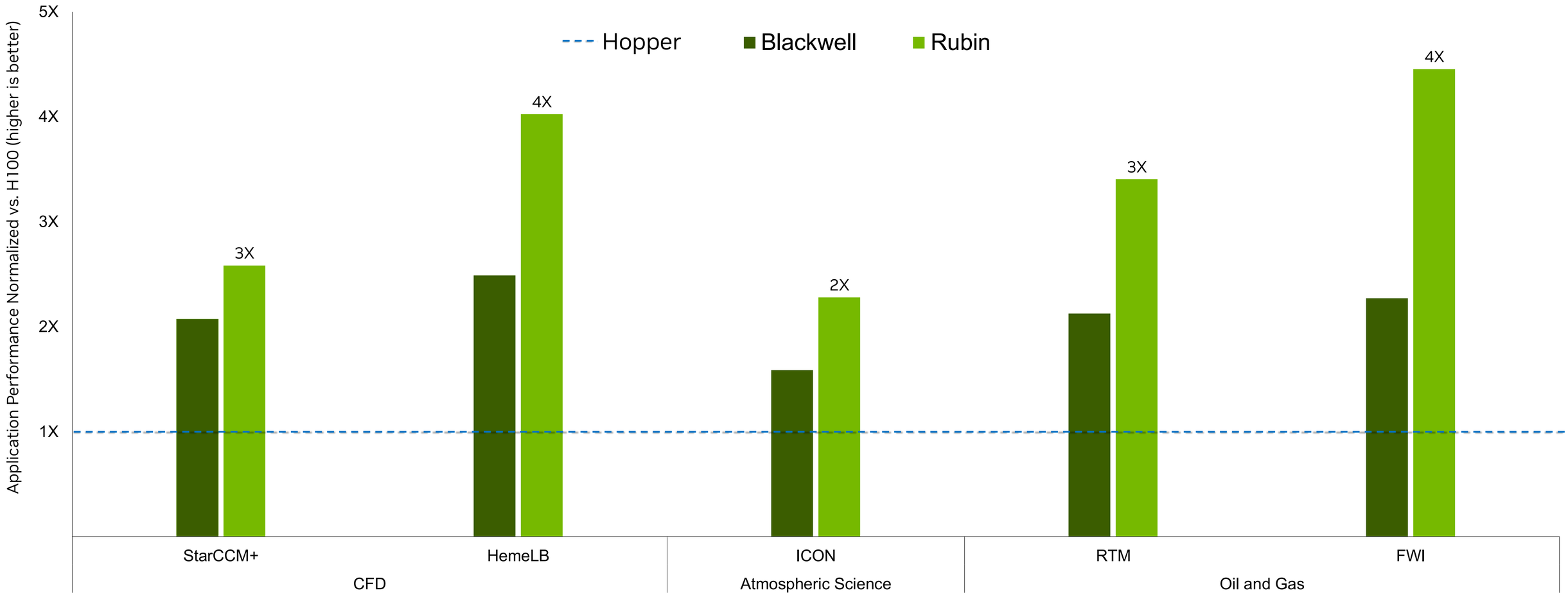
64-bit precision plays a necessary role

Application Performance

	Ampere	Hopper	Blackwell	Rubin
				
Native FP64	✓	✓	✓	✓

Up to 4X Performance for Fluid Dynamics and Geo Science

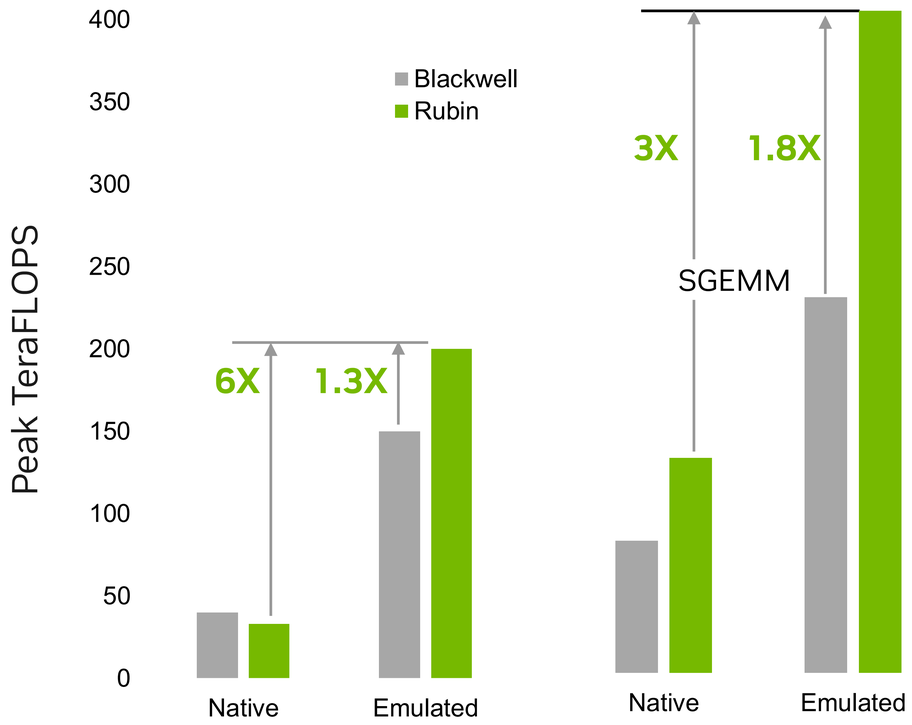
vs. Hopper



Projected performance on Rubin. Subject to Change. Power assumptions Hopper 700W, Blackwell 1200W, Rubin 1800W

Zero Code Change Application Performance For Dense Matrix Multiply

Up to 6X additional performance by enabling emulation

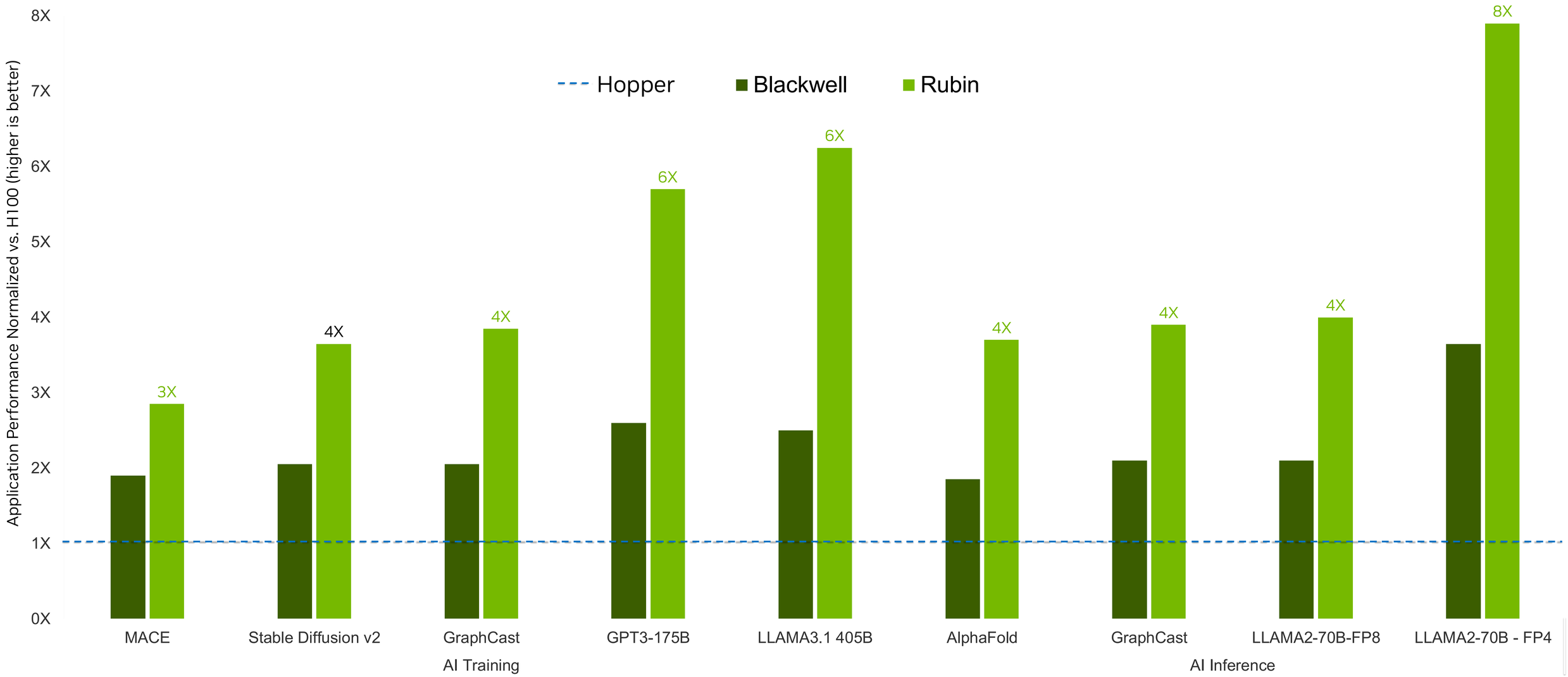


Reliable	Accuracy guaranteed. Incorrect results are a bug that NVIDIA will fix.
Performant	Up to 4X on Blackwell and 6X on Rubin with more Improvements coming with CUDA-X.
Available	You have access to this today. Try it.

NVIDIA.

Up to 8X more Performance for AI training and inference

vs. Hopper

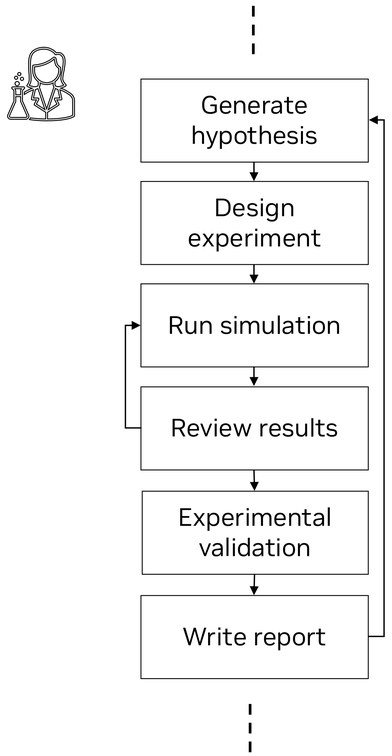


Projected performance on Rubin. Subject to Change. Power assumptions Hopper 700W, Blackwell 1200W, Rubin 1800W

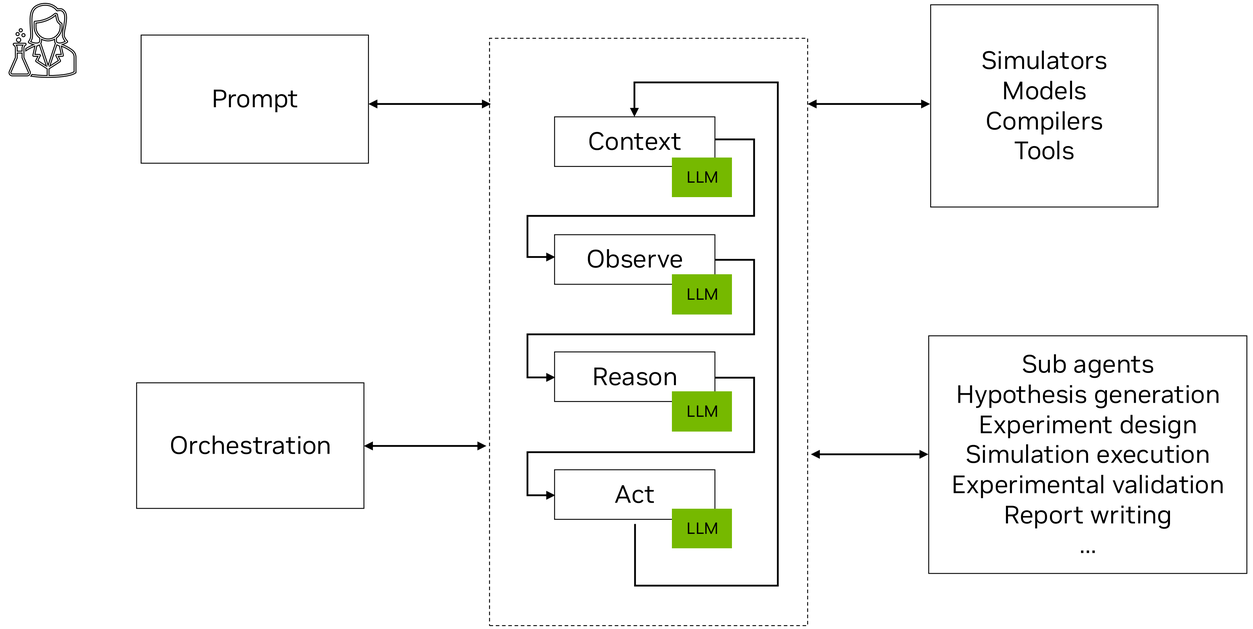
Agentic AI for Science

Agentic AI Expands Reach of Scientific Computing

Expanding capabilities, accessibility and applications leading to new discoveries



Traditional Workflow

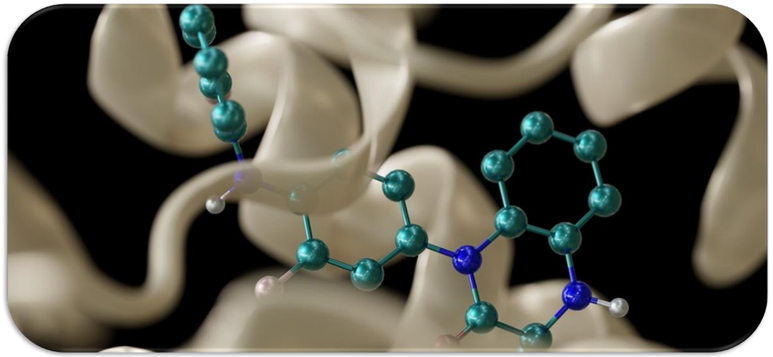


Agentic Workflow

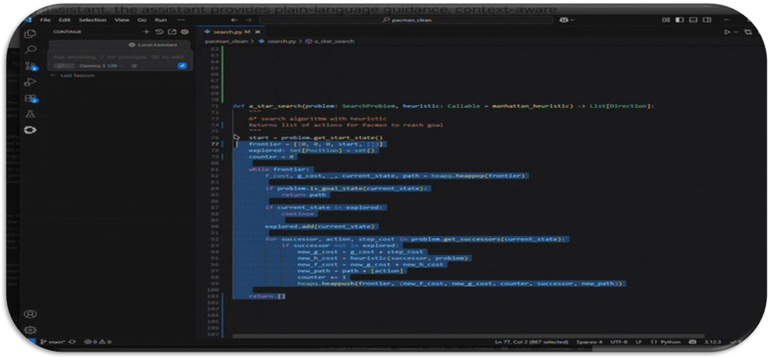
NVIDIA

New Trend: AI Scientists

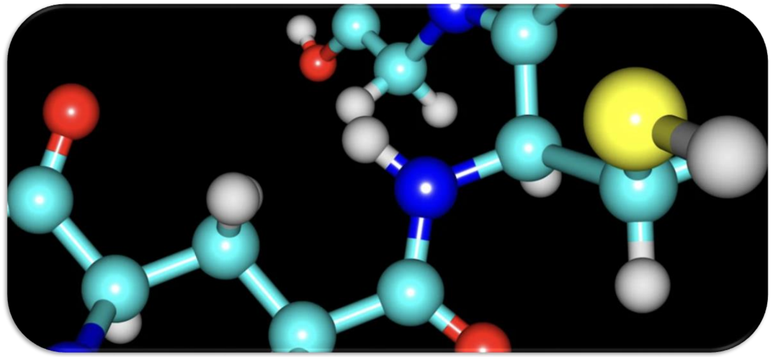
AI Scientists emerging across every domain



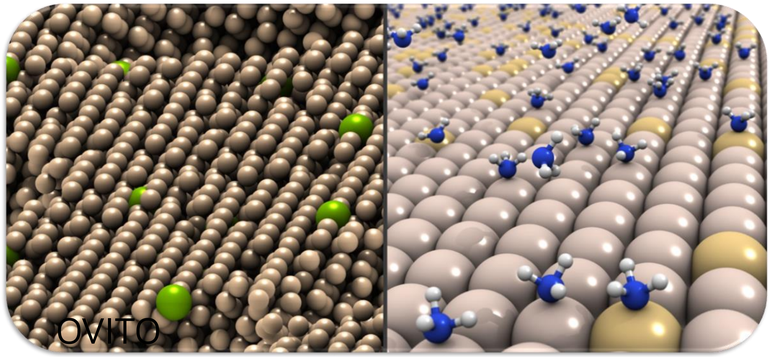
Automating Drug Discovery



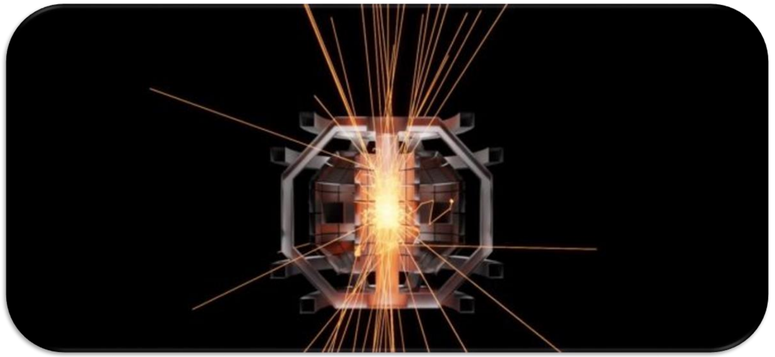
Automating Coding



Designing New Molecules



Materials Candidate Generation



Fusion Hypothesis Generation

Optimizing Clinical Workflows

Nemotron 3 Open Models

Frontier Smart. Faster. More Cost Efficient

Ultra

550B-A55B

Frontier Smart

5X Faster

30% Lower Compute Cost

Super

120B-A12B

4X Throughput with High Accuracy

Nano

30B-A3B

60% Savings with Thinking Budget

Open Datasets

Pre-Training Tokens, Post-Training Samples, RL Tasks

NeMo Gym

RL Environments & Skills

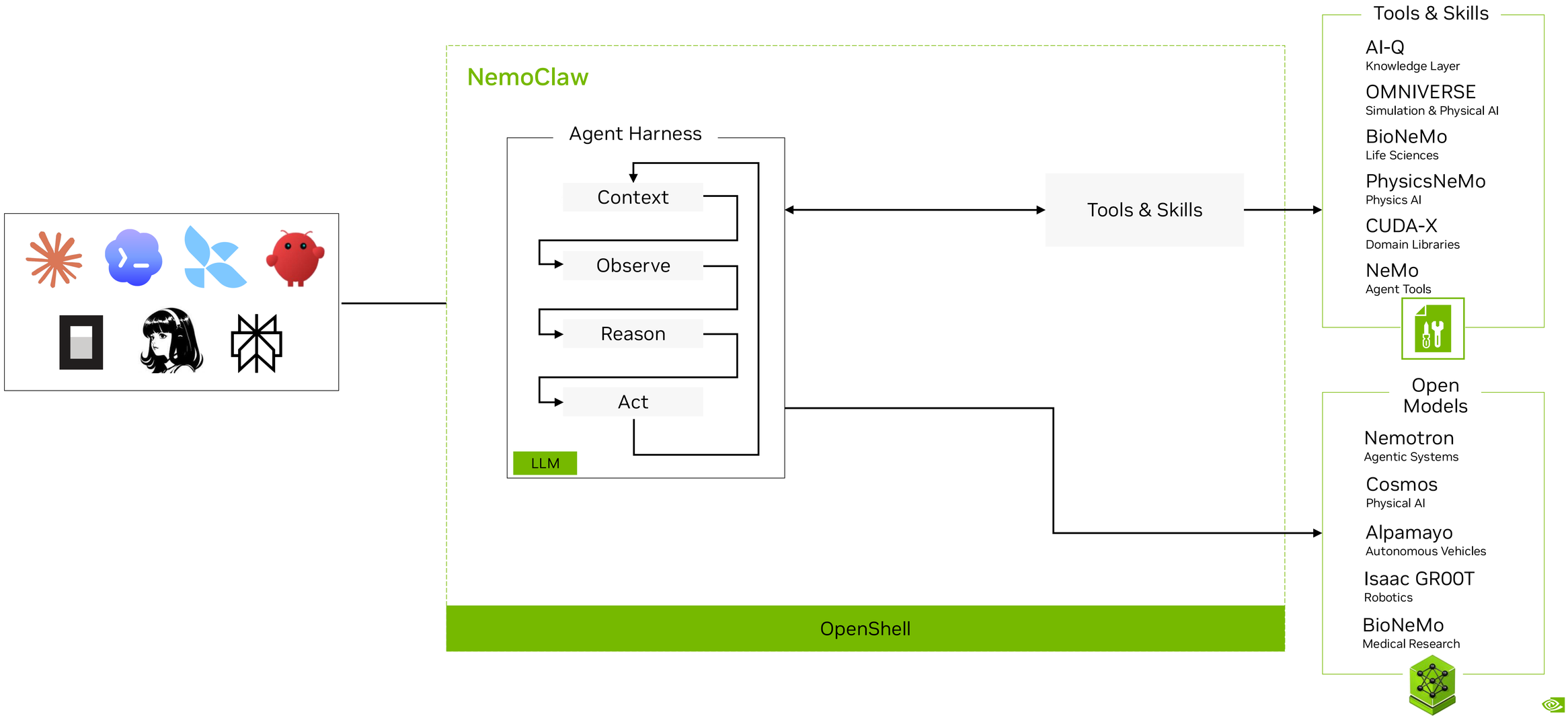
Open Research

Papers, Samples

NVIDIA

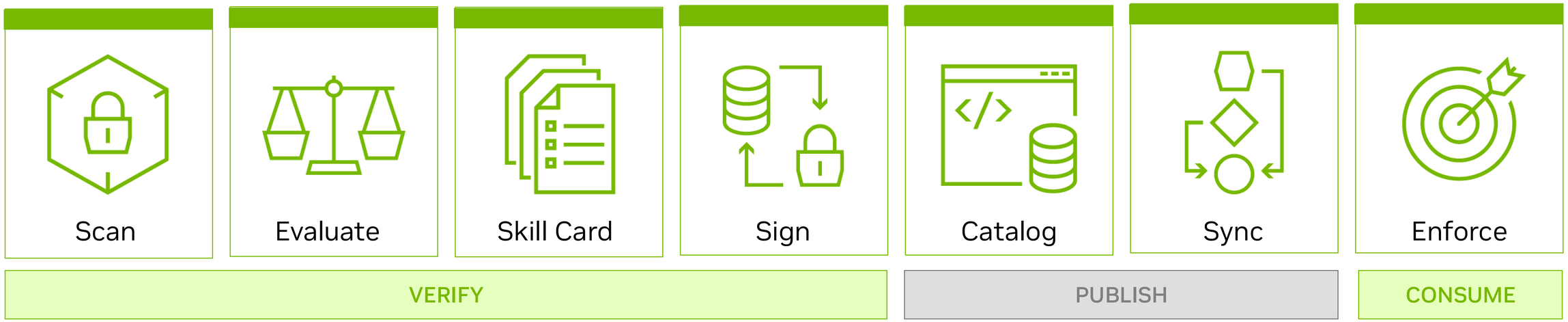
NVIDIA Agent Toolkit for Enterprise AI

Building Blocks for Specialized Agents



NVIDIA Libraries Available as Skills for Agents

github.com/NVIDIA/skills



Available as plugins with your preferred agent:

NVIDIA

What should we work on next?

Auto-review

5.5 Medium

Components in NVIDIA Agent Toolkit

Open stack for building and running enterprise AI agents

OPEN MODELS

Nemotron

Open models for reasoning, safety, and multimodal intelligence.

NIM

Optimized inference microservices that expose models through standard APIs.

Dynamo

Distributed inference orchestration for efficient, scalable model serving.

AGENT TOOLS & SKILLS

NeMo RL & NeMo Gym

Reinforcement learning and scalable environments for post-training, rollouts, and evaluating agent behavior.

NeMo Relay

Control and observability layer for agent runs, tool calls and LLM calls across existing frameworks.

NeMo Switchyard

Task-aware model routing across endpoints to optimize quality, latency and cost.

Omniverse

Libraries for physical AI, simulation, and 3D workflow automation.

BioNeMo

Libraries for life sciences, enabling agents for drug discovery, protein structure prediction, and genomics workflows.

PhysicsNeMo

Open framework for scalable physics AI models for science and engineering.

BLUEPRINTS

NemoClaw

Blueprints for custom, super agents that turn your company's proprietary intelligence, data and domain expertise into action.

AI-Q

Enterprise AI knowledge engine that connects agents to structured and unstructured data – making it understandable, and predictive.

SECURE RUNTIME

OpenShell

Policy-based controls for privacy, networks, tools, files, and code execution.

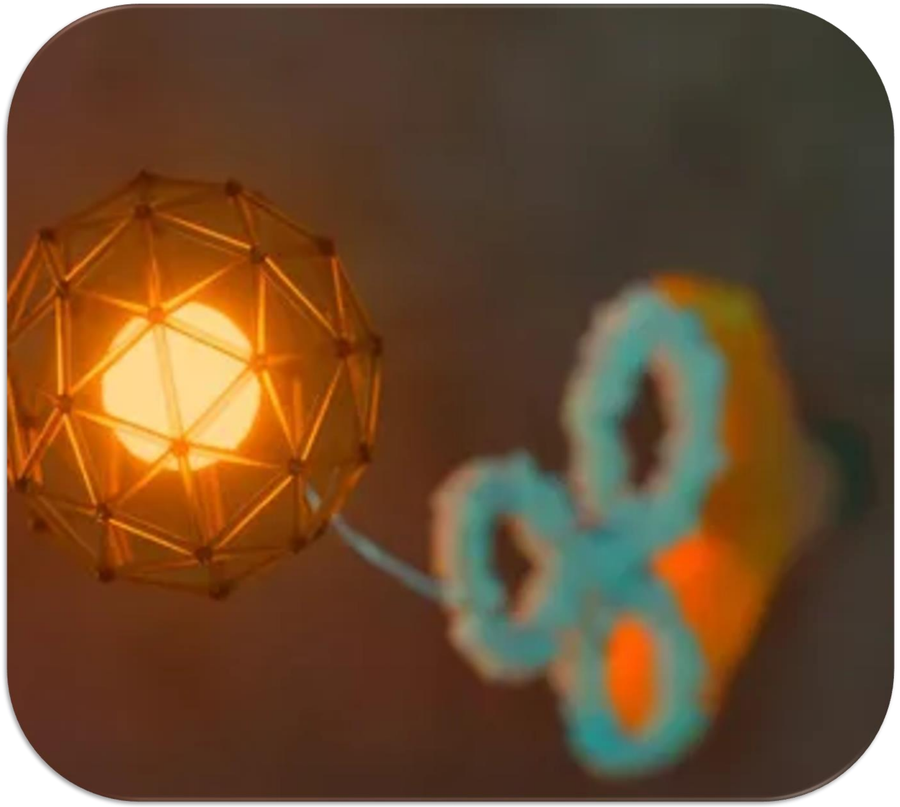
NVIDIA

AI Scientists for Fusion Research and Cancer Treatment

LANL powering agentic AI for science with Llama Nemotron and GenMol

Hypothesis Generation for ICF Fusion

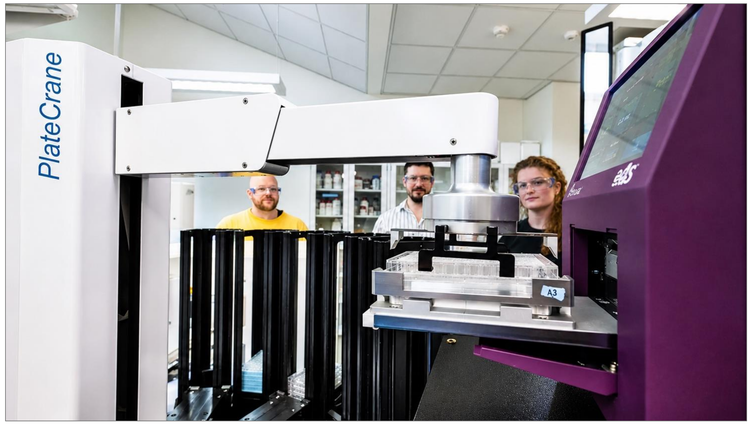
Llama Nemotron, Nemo Frameworks



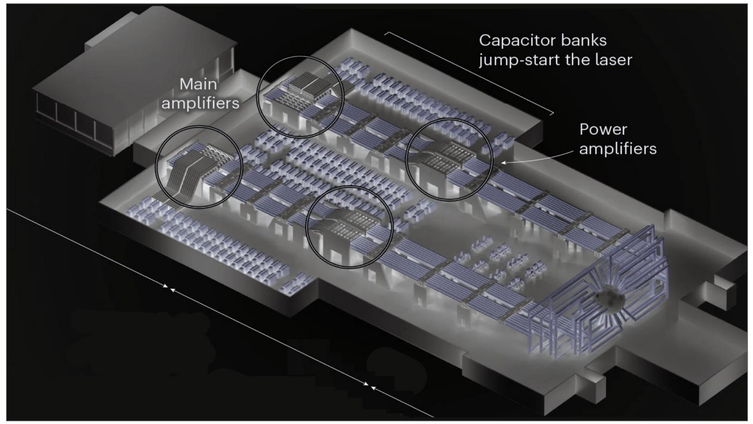
Chelator Molecule Design

Llama Nemotron, GenMol, Venado

Agentic AI is Driving the Next Wave of Science and Engineering



ROSA: Robotic Scientific Assistant
Argonne National Lab

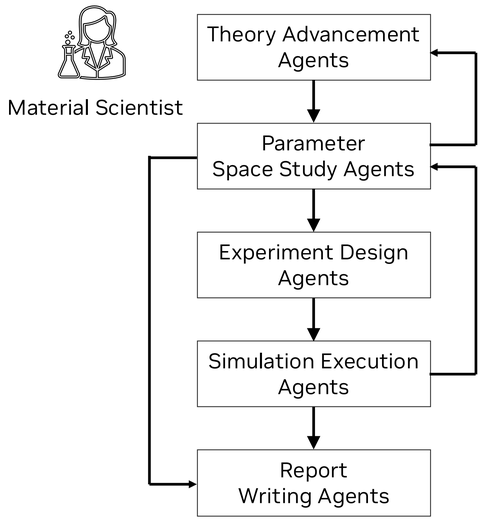


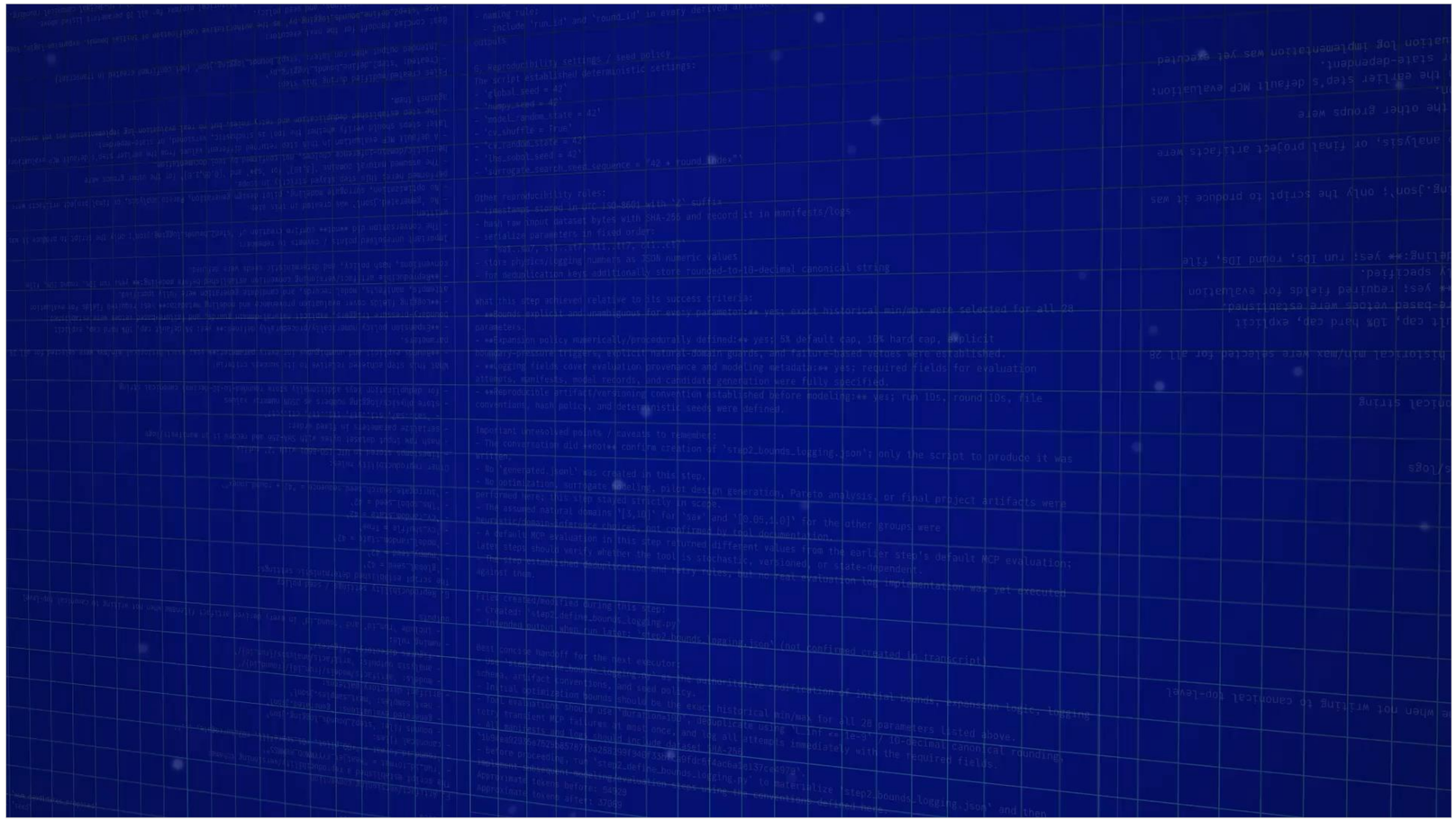
URSA: Universal Research Scientific Agent
Los Almos National Lab

Automated Quantum Calibration
Aegiq

LANL, University of Michigan, and Open AI Demonstrate URSA - Agentic AI for Material Discovery

AI framework helps scientists plan, run, and analyze experiments connected to scientific foundation model (SciFM) surrogates





CUDA-X Agentic Skills for Science

The Foundation of Modern Science

7M
Developers

600M
Downloads

6K
Accelerated Applications

1K
SDKs and Models

cuLitho
Computational Lithography

NemoClaw
OpenShell
AI-Q
VSS
NeMo
Agentic AI

Megatron
Dynamo
cuDNN
CUTLASS
Voice Agent
Deep Learning and Inference

cuDSS
cuSPARSE
cuFFT
AmgX
Computer-Aided Engineering

Warp
PhysicsNeMo
AI Physics

cuOpt
cuDF
cuML
cuVS
Retriever
cuPhoton
Data Science and Processing

Newton
IsaacLab
Robotics

Omniverse
Real-Time Digital Twins

NVFlare
Federated Learning

cuEST
cuEquivariance
cuTensor
ALCHEMI
Quantum Chemistry

cuQuantum
CUDA-Q
Quantum Computing

Earth-2
Weather Analytics

DOCA Memos
Context Memory Storage

MONAI
Medical Imaging

Parabricks
Genomics

Aerial
Sionna
Holoscan

DAQIRI
Signal Processing

Every Industry | One Architecture | CUDA Everywhere

Vera CPU

NVIDIA Vera CPU To Boost Agentic Science

2,300 stand-alone Vera CPUs
2,160 Rubin GPUs + 1,080 Vera CPUs
Quantum-X800 InfiniBand

1,150 stand-alone Vera CPUs
576 Rubin GPUs + 288 Vera CPUs
Quantum-X800 InfiniBand

Agentic Tool Performance

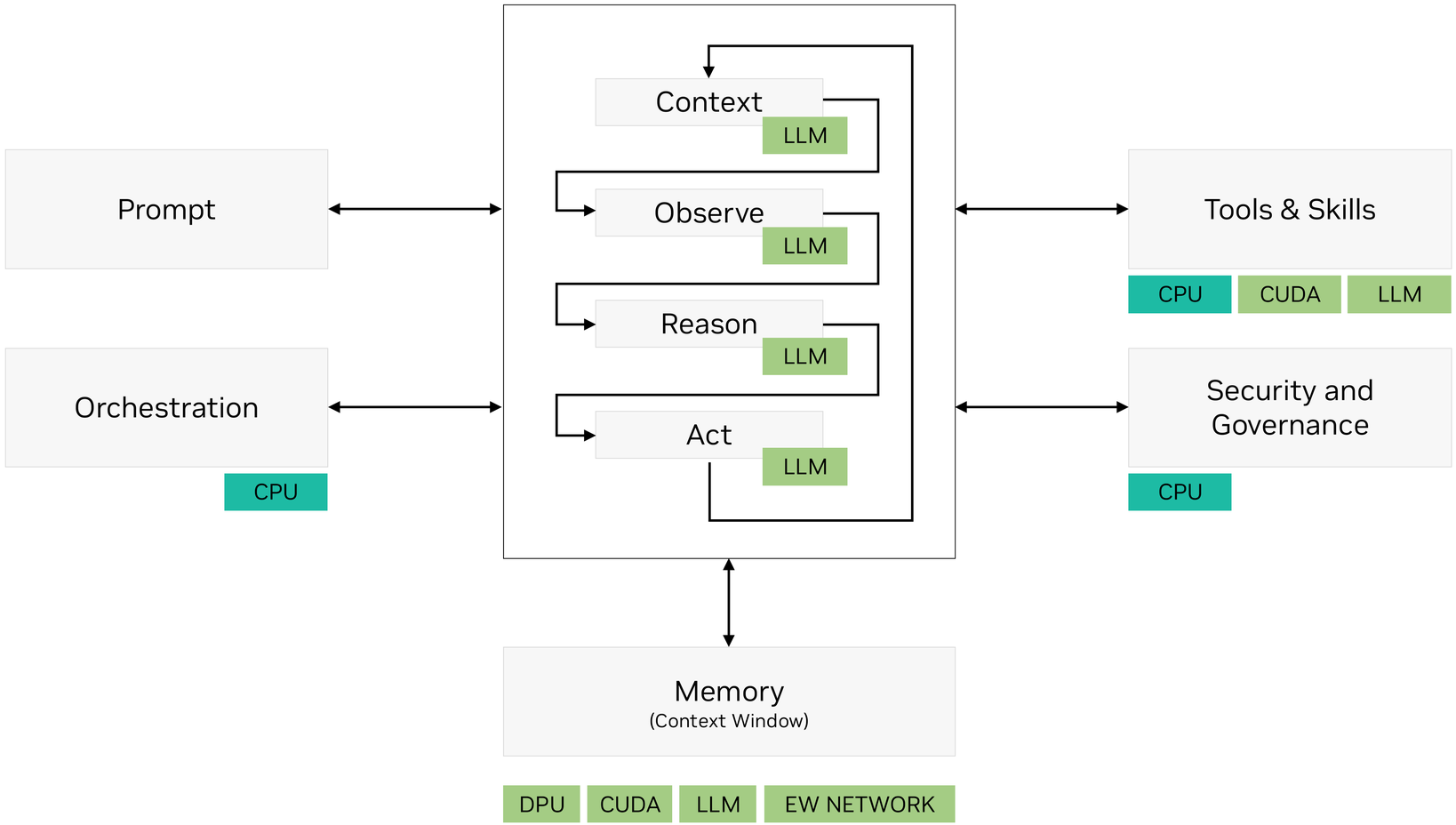
1.7x
Code compilation

1.7x
Code analysis

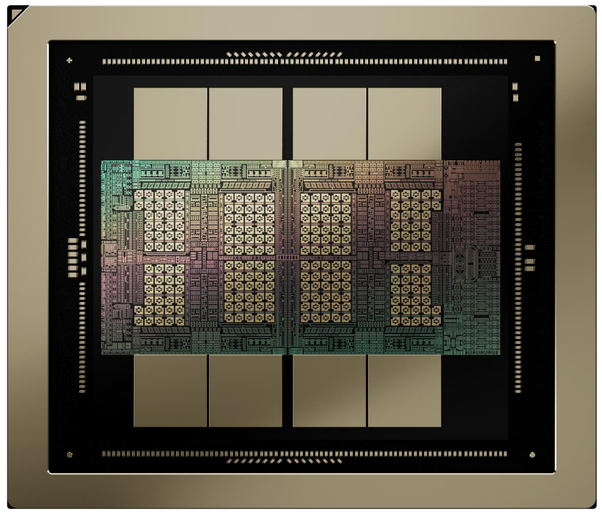
1.9x
Python

NVIDIA

Agentic Workflow Demands New AI Computing Platform

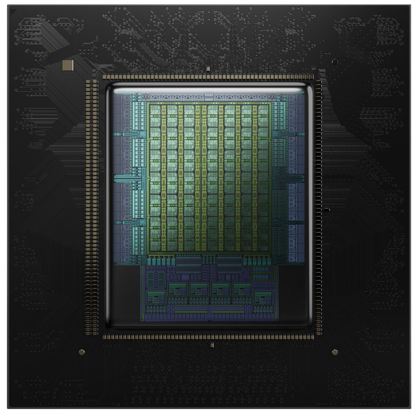


Agentic AI and Reinforcement Learning Need Fast CPUs



Tokens
Code and Queries

Evaluations
Results and Data



Model / Agent
Rubin GPU

Reinforcement Learning
Training
Agent Inference

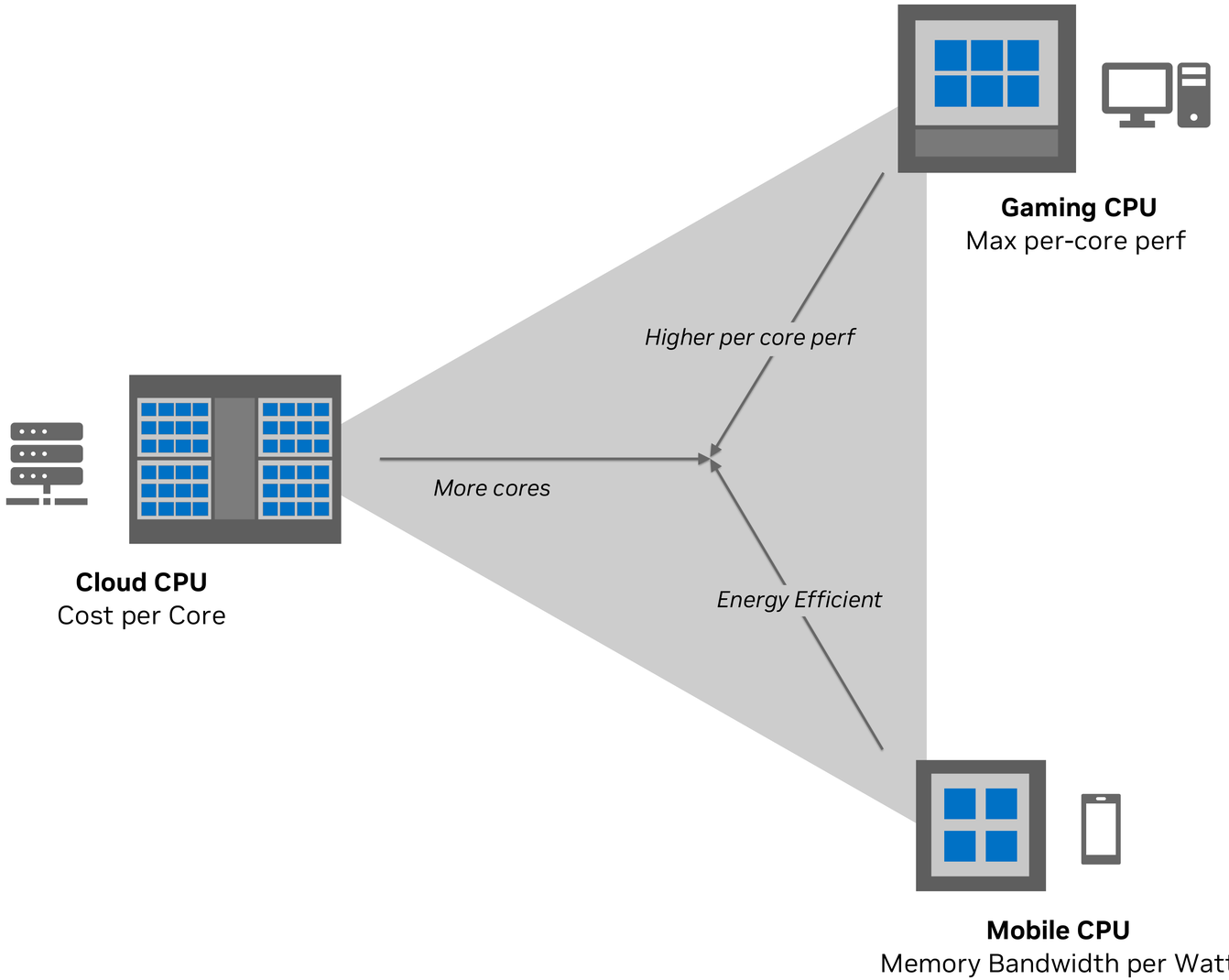
Sandbox Execution
Vera CPU

Tool Calls, Simulation & Tests
Search, Query, Transform
API and Browser

NVIDIA

Agentic AI Needs a New CPU

Existing CPUs not optimized for agentic AI workloads



NVIDIA

Announcing NVIDIA Unveils Vera, the CPU for Agents

High-Performance Custom Olympus Cores

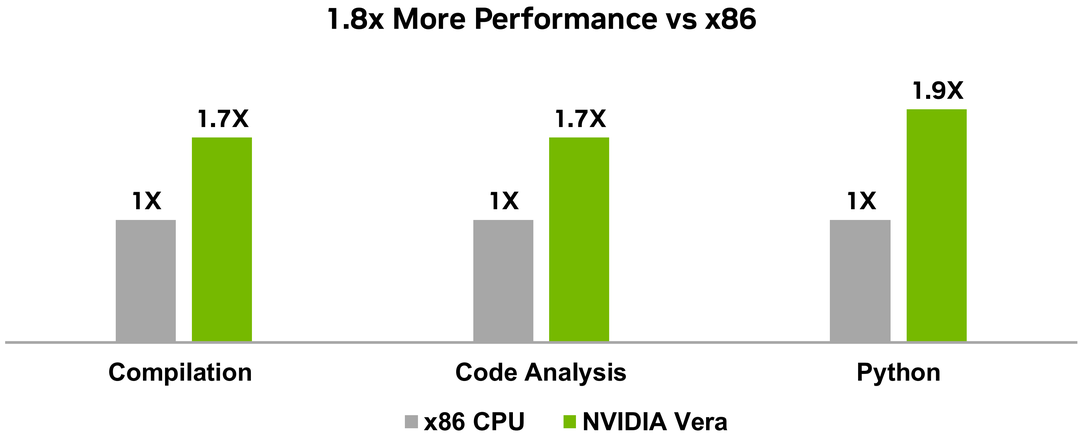
50% Higher IPC

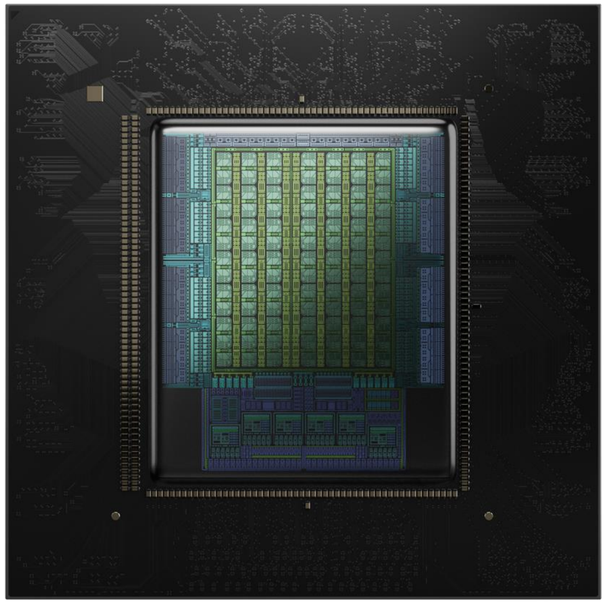
Massive Memory Bandwidth

1.2 TB/s LPDDR5X BW

Fast On-Chip Fabric

3.4 TB/s BW



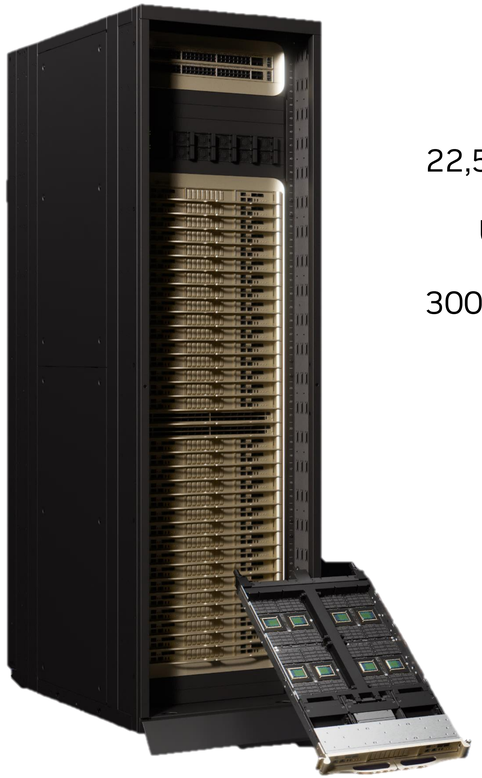


88 NVIDIA Custom Olympus Cores
176 Threads NVIDIA Spatial Multi-Threading
1.2 TB/s LPDDR5X
1.8 TB/s NVLink-C2C
1.8x agentic AI performance

NVIDIA

NVIDIA Vera CPU: Purpose-Built for Agentic AI and High-Performance Science

The Custom Olympus Core — 88 Cores Engineered for Scientific Simulation and AI Tool Loops



Vera CPU Rack
Liquid Cooled MGX Rack

256 Vera CPUs

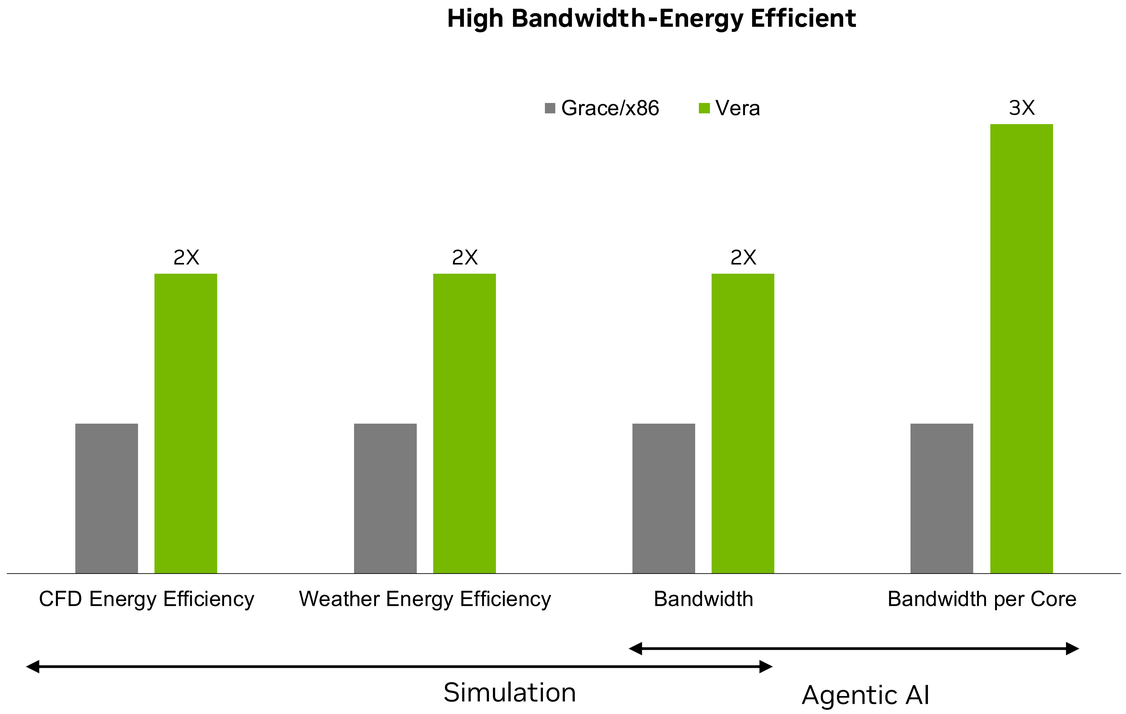
22,528 Cores | 45,056 Threads

Up to 400 TB LPDDR5X

300 TB/s LPDDR5X Bandwidth

64 BlueFiled-4 DPUs

Vera CPU Servers
2S and 1S Air Cooled



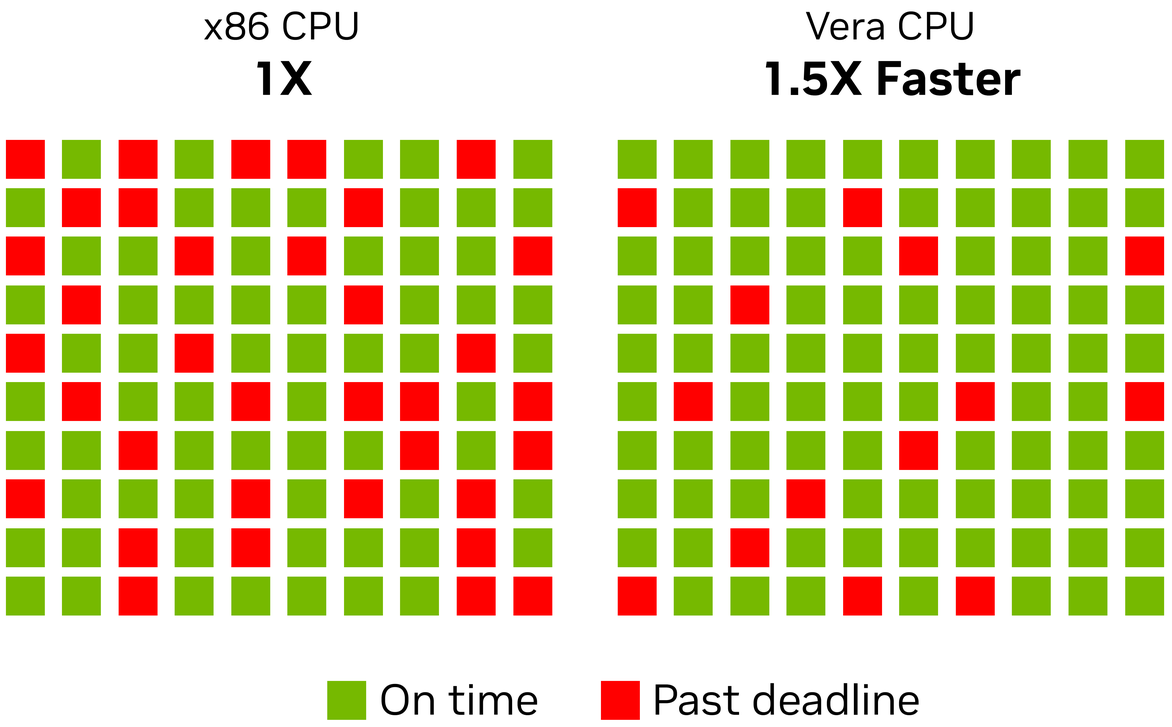
- Agentic and Reinforcement Learning Workflow:
  - Tool Calls, Simulation & Tests
  - SQL Queries, Compiles Code, Python Execution

2X Memory Bandwidth v x86. 2X Memory per core v x86.2X CFD Perf/Watt v Grace. 2X Weather Perf/Watt v Grace.
CFD: OpenFOAM. NAS Parallel Benchmark. Weather: WRF, ICON, IFS, FEMOS. All performance results are estimates. Results subject to change.

NVIDIA

NVIDIA Olympus Cores Increase Agentic Throughput

Faster cores complete more sandbox work inside each AI loop

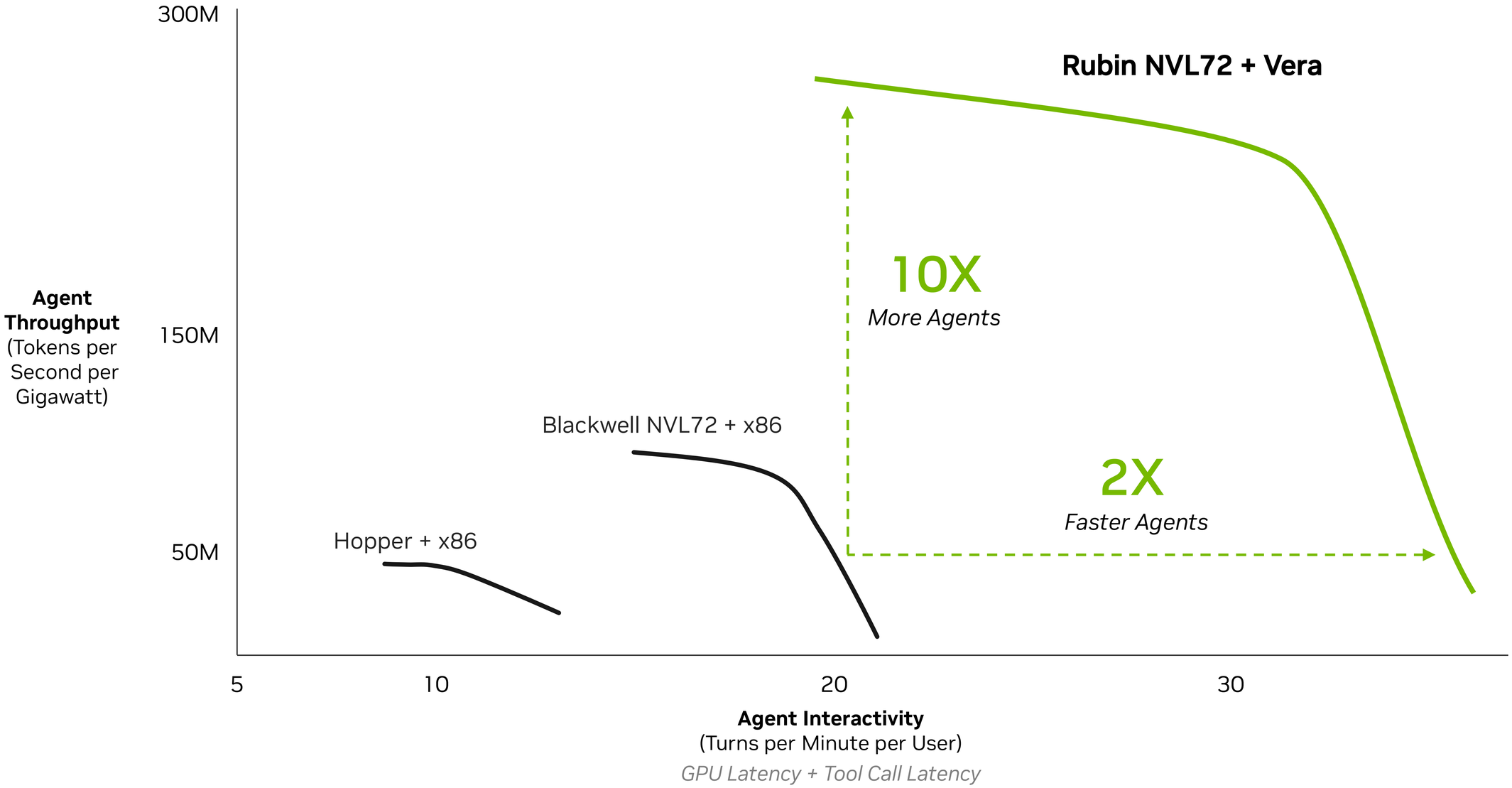


Same time budget. More Vera sandboxes finish before the next AI loop.

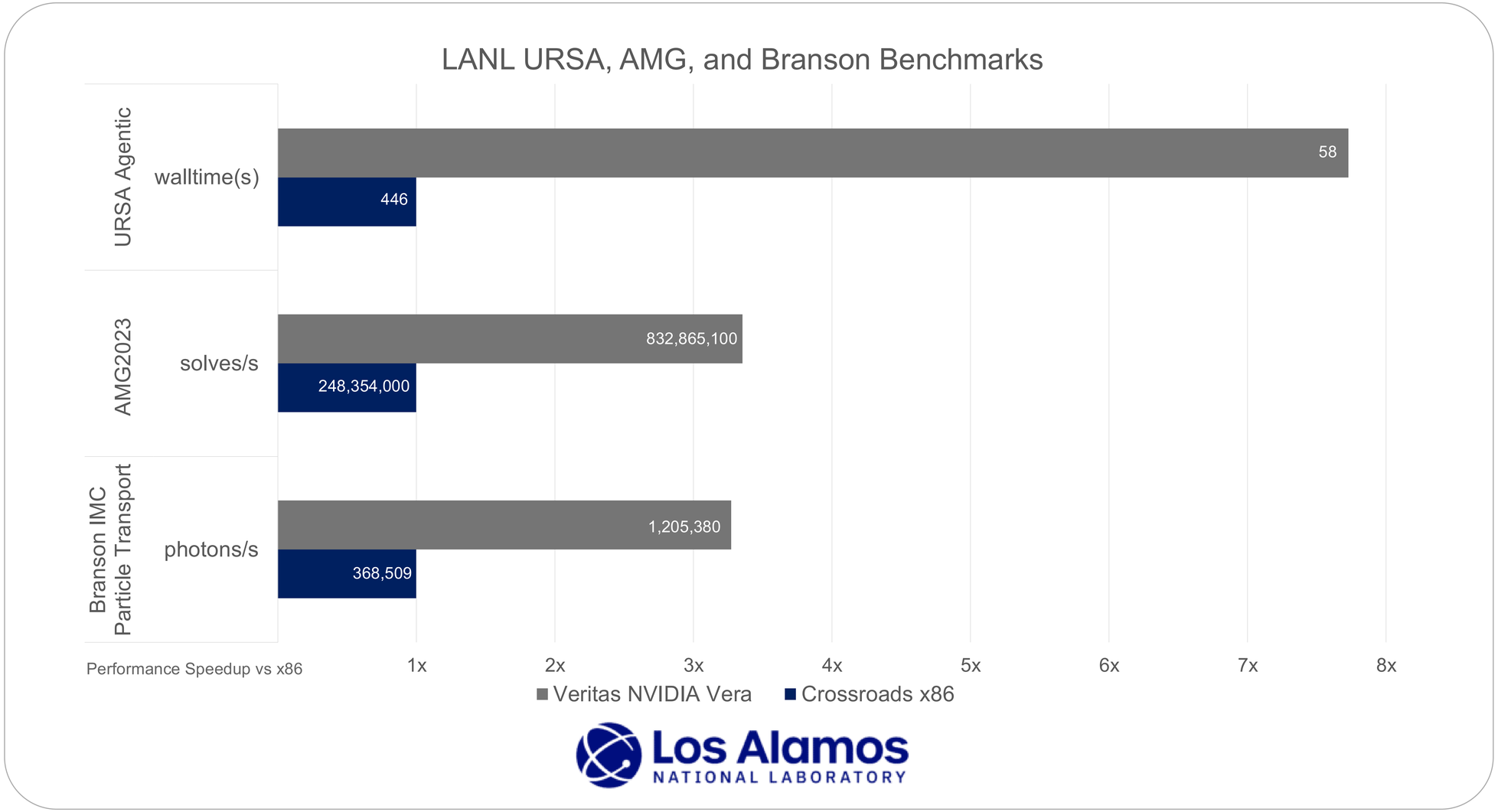
Built for Branch-heavy Agentic Code

- 10-wide decode
- Neural branch predictor
- Large out-of-order engine
- Advanced prefetchers
- Graph data analytics prefetch engine

Extreme Co-Design Drives Agentic Performance



Vera CPU 3-7x Faster for Agentic and Simulation Workloads



High Performance, Energy-Efficient Systems in TOP500/GREEN500

NVIDIA powers 81% of Top500 and sweeps Green500 for performance and energy efficiency

Powering Worlds Fastest, Greenest Supercomputers

TOP500
The platform of choice

81%
use NVIDIA
(+17 vs SC25)

238
GPU systems
record high

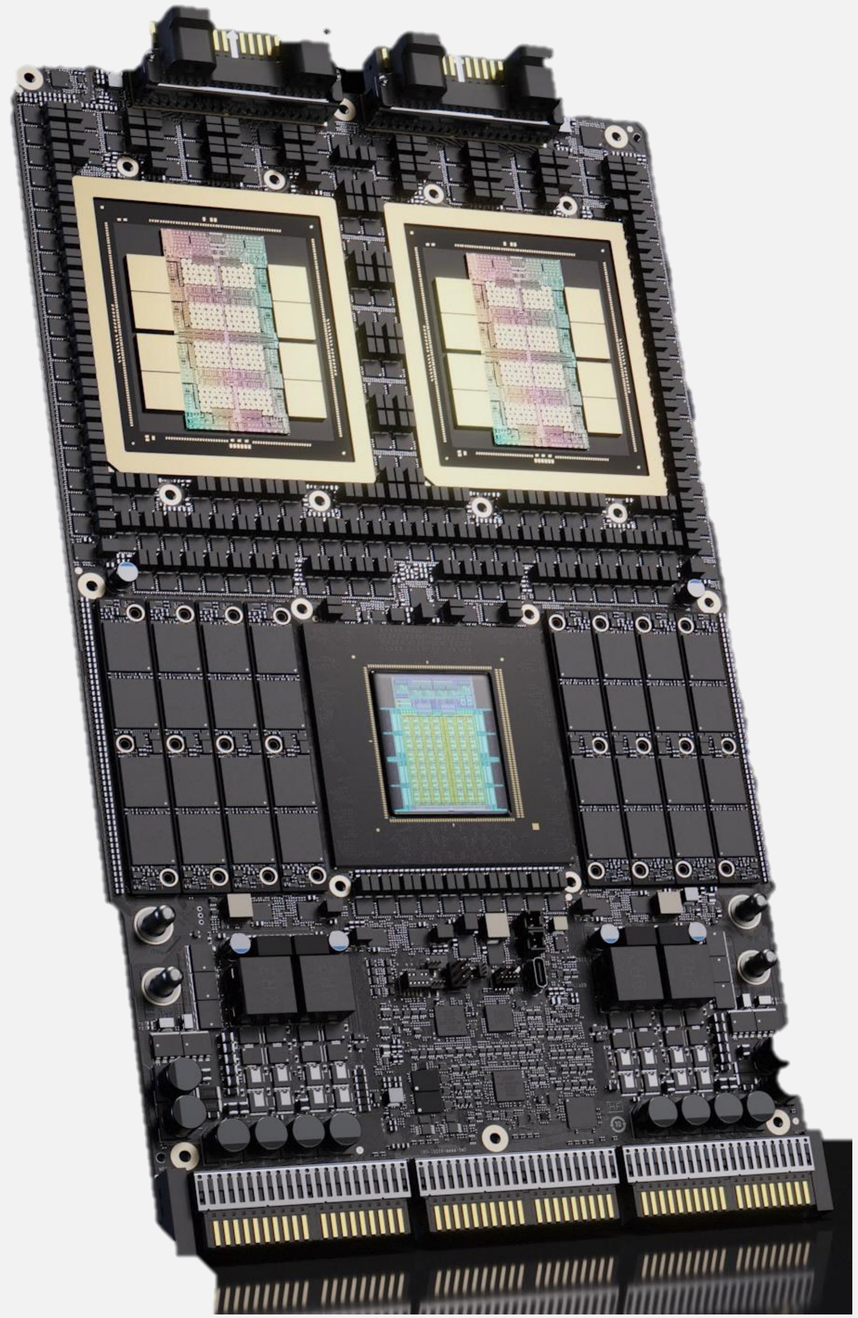
89%
Of new systems
choose NVIDIA

Green500
Energy Efficiency

#1
Green500
GH200 73GF/W

8 of the
Top 8 powered by
NVIDIA Hopper

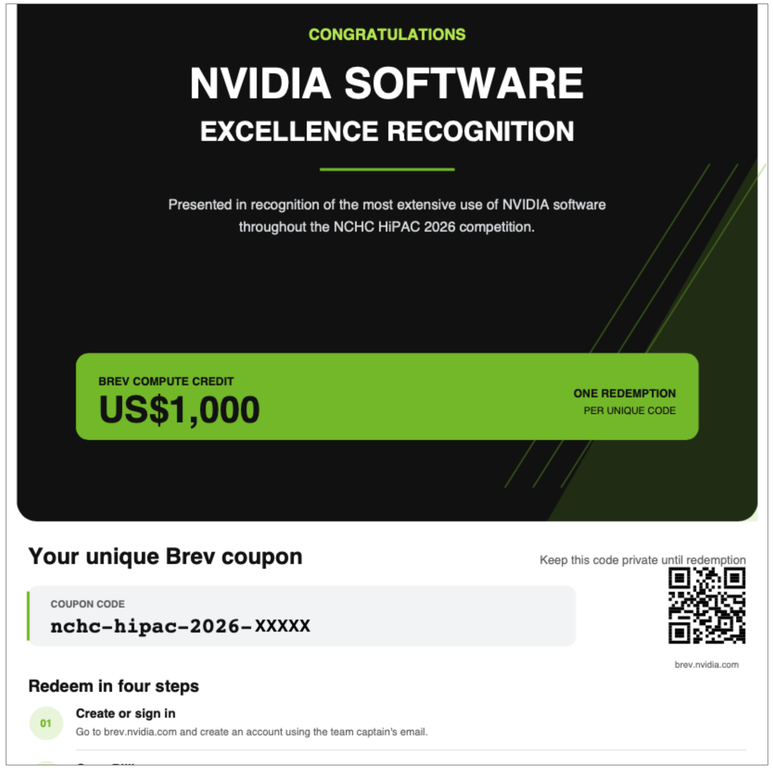
68%
of the top 50
Green500 are
NVIDIA



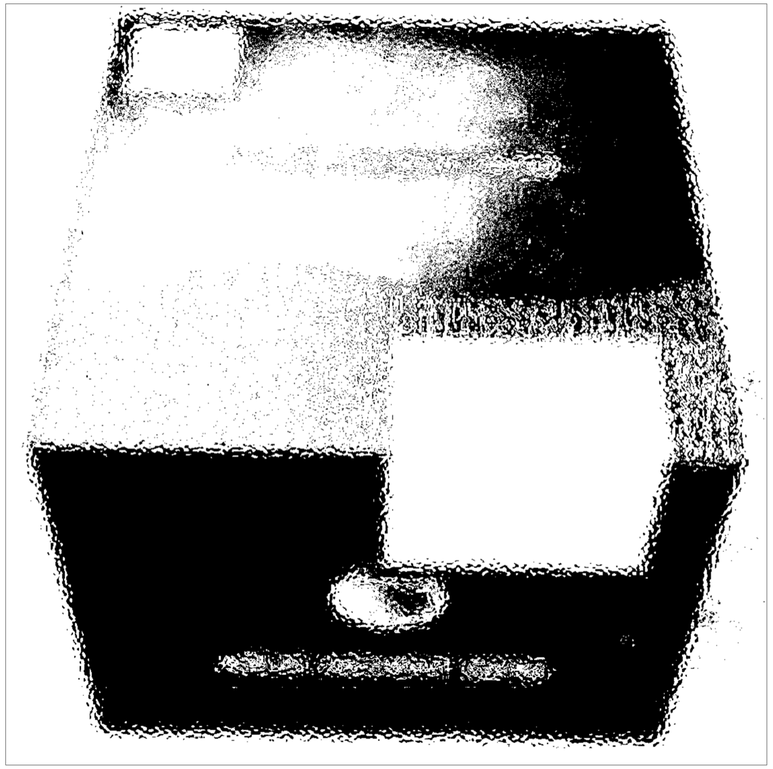
NVIDIA Special Awards

Announcing NVIDIA Special Awards for NCHC HiPAC 2026

NVIDIA LOGO GTC TOTE
FOR EVERYONE

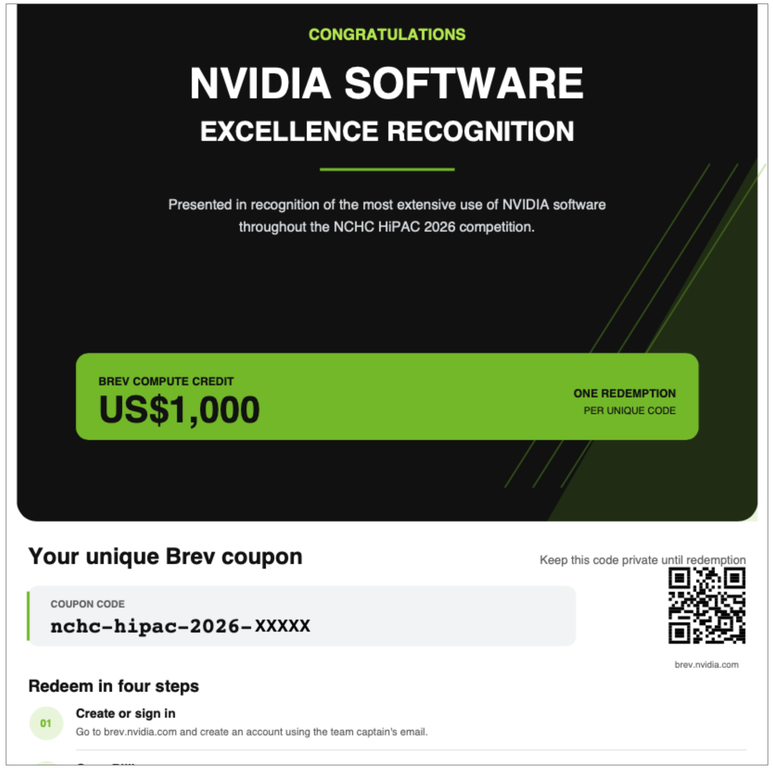


NVIDIA SOFTWARE EXELLENCE
RECOGNITION FOR 3 TEAMs



Announcing NVIDIA Special Awards for NCHC HiPAC 2026

NVIDIA LOGO GTC TOTE
FOR EVERYONE



NVIDIA SOFTWARE EXELLENCE
RECOGNITION FOR 3 TEAMs

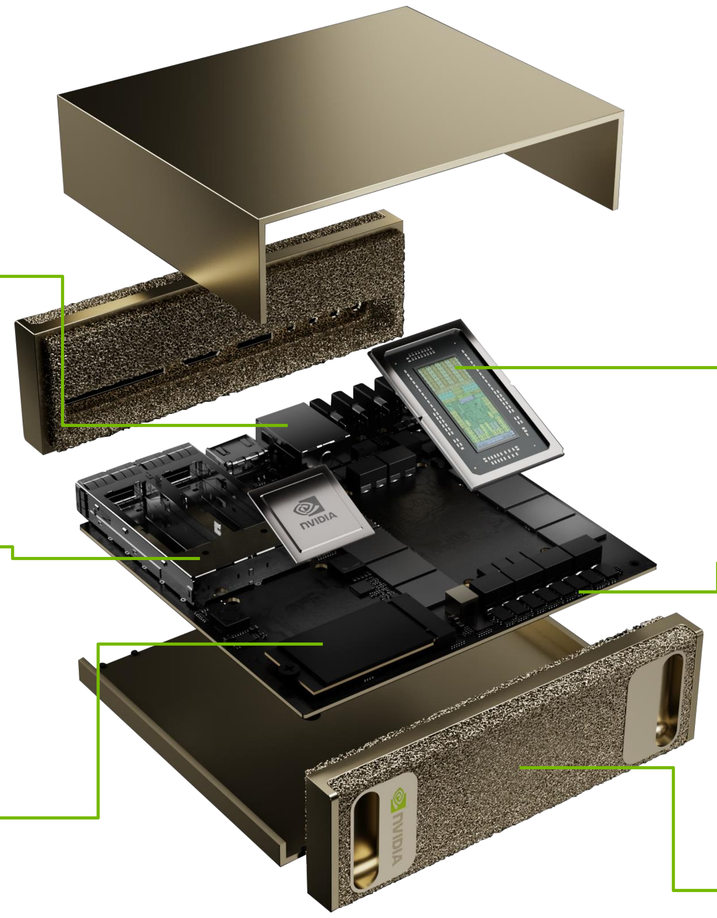
ASUS GX10
FOR #1 WINNER

Key Features

Connectivity
10Gb Ethernet
Wi-Fi 7, Bluetooth, USB

ConnectX 7
Cluster GB10 systems
200Gb Ethernet x 2

1 - 4TB SSD Storage



NVIDIA GB10 Superchip
Latest Gen Blackwell GPU
20 Arm Cores
1 Petaflop FP4 AI Compute

High-Bandwidth Unified Memory
128 GB Low Power LPDDR5x

NVIDIA AI Software
Blueprints, NIMs, NEMO,
& Toolkits etc.

NVIDIA DGX Base OS
Built on Ubuntu

NCHC NVIDIA Joint-Lab

NCHC x NVIDIA Workshops/Bootcamps/Hackathons

Since 2020

Events

56+

Developers

2800+

Bootcamps are published on NCHC Youtube Channel

國網中心NCHC

@nchc3005 · 1.97K subscribers · 213 videos

國家高速網路與計算中心 ...more

nchc.org.tw and 2 more links

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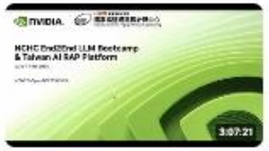

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Workshops and Bootcamps are Open-Sourced

https://github.com/nqobu/nvidia

Repository for NCHC-NVIDIA Joint Lab

- tutorials - NVIDIA online courses/tutorials in AI/HPC
- 20210412 - NVIDIA Techniques Sharing 2021
- 20210706 - NCHC Techniques Sharing 2021
- 20211202 - NVIDIA Techniques Update 2021
- 20211221 - AI+HPC: 利用 NVIDIA Modulus 實踐 PINN 於物理模擬
- 20220415 - NVIDIA Techniques Update 2022
- 20220530 - NCHC-NVIDIA Techniques Sharing 2022
- 20220629 - NCHC PINN 與 NVIDIA Modulus 實作訓練營
- 20221111 - NCHC Quantum Computing Workshop / 量子計算模擬
- 20230413 - NVIDIA Techniques Sharing 2023
- 20230517 - NVIDIA Techniques Briefing: NVIDIA Federated Learning
- 20230525 - AI for Science: NVIDIA Modulus 及 NVIDIA Omniverse 實作
- 20230727 - NCHC N-Way to GPU Programming Bootcamp / 多 GPU 程式設計訓練課程
- 20230821 - NVIDIA Techniques Salon 2023: Programming the NVIDIA Superchip
- 20231207 - NCHC Open Hackathon 2023
- 20240410 - NCHC Quantum Computing Bootcamp 2024 / NVIDIA CUDA-Q and cuQuantum
- 20240506 - AI for Science: NVIDIA Modulus, NVIDIA Omniverse, and NVIDIA Earth-2
- 20240508 - NCHC Techniques Sharing 2024
- 20240626 - NCHC AI for Science Bootcamp 2024 / NVIDIA Modulus 物理模擬計算
- 20240806 - NCHC End-to-end LLM Bootcamp 2024 / NVIDIA NeMo 大型語言模型框架
- 20240924 - NCHC N-Way Bootcamp 2024 / NVIDIA GPU 加速運算
- 20241114 - NVIDIA Earth-2 Overview
- 20241129 - NCHC × NTU - NVIDIA BioNeMo Protein Design Workshop 2024
- 20241204 - NCHC Open Hackathons 2024
- 20250218 - NCHC Grace Workshop 2025
- 20250415 - NCHC N-Way Bootcamp 2025 / NVIDIA GPU 加速運算訓練課程
- 20250617 - NCHC LLM Bootcamp 2025 / 大語言模型訓練課程
- 20250723 - NCHC CUDA-Q Bootcamp 2025 / 量子計算模擬訓練課程
- 20250904 - NCHC × NCKU – Digital Biology Bootcamp / 數位生物技術訓練營
- 20251110 - NCHC Quantum-HPC Tech Salon
- 20251112 - NCHC Open Hackathons 2025
- 20260310 - NCHC AI Powered Physics Bootcamp, 2026
- 20260520 - NCHC LLM & Agentic AI Bootcamp, 2026

NVIDIA

關鍵議題研究中心
Research Center for Critical Issues

SINICA NCHC
CUDA-Q
AGENTIC CODING BOOTCAMP

Hands-on training on NVIDIA Quantum Platform, hybrid quantum-classical workflows, and agentic AI for real quantum hardware

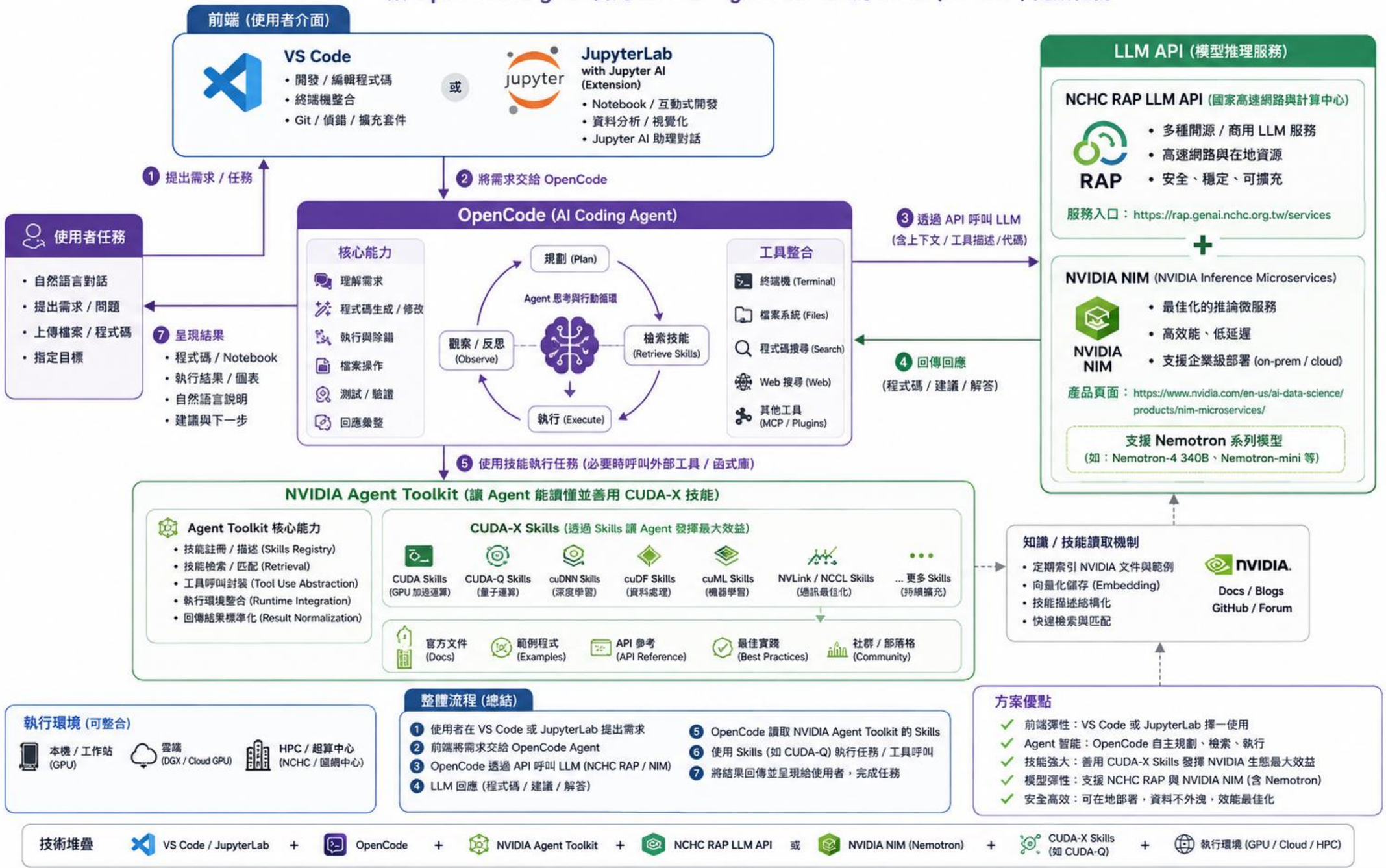
2026/8/7（五）
9:30-16:20

中央研究院南院
關鍵議題研究中心1F演講廳

09:30-10:00	Registration & Setup	
10:00-10:15	Opening	Academia Sinica | 陳啟東教授
10:15-10:45	NVIDIA Quantum Platform Overview: CUDA-Q & NVIDIA Ising	NVIDIA | Pika Wang
10:45-11:15	NVIDIA CUDA-Q Environment Setup	
11:15-12:00	Basic CUDA-Q Hands-on + Ising Calibration Playground (NIM)	
12:00-13:00	Lunch Break	
13:00-14:00	Superconducting QPU Introduction & Agentic Calibration	Academia Sinica | 陳啟東教授 NCHC | 李泰岳博士, 何明鋼博士
14:00-14:10	NCHC Tech Sharing	NCHC | 李泰岳博士
14:10-14:20	Agentic Coding with NCHC RAP Service	NVIDIA | Jay Chen
14:20-14:40	Tea Break	
14:40-15:20	Advanced CUDA-Q Hands-on with RAP	NVIDIA | Pika Wang
15:20-16:00	Bring-Your-Own-Code with CUDA-Q Agentic Programming	
16:00-16:20	Q&A + Open Discussion	

OpenCode + VSCode / JupyterLab + NCHC RAP LLM API / NVIDIA NIM + Nemotron

讓 OpenCode Agent 善用 NVIDIA Agent Toolkit 的 Skills (CUDA-X) 完成任務



2026 NCHC Open Hackathon

Connect NVIDIA Experts	Accelerate Workloads with **AI Agents**	Celebrate Together
Developers & Mentors Apps & Acceleration	Speedup Energy Efficient	Publication, Co-Paper Blogs and Talks

Please ack NCHC and Mentors if any publications.

OpenACC More Science, Less Programming

OPEN HACKATHONS

國家實驗研究院 國家高速網路與計算中心 National Center for High-performance Computing

NVIDIA

Publications

https://www.nchc.org.tw/Message?itemid=1008&mid=165

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最新文章 AI 人工智慧 AI 機器人 智慧製造 新零售 金融科技 資安 主題特展 TO Highlight 全新一週

AI 人工智慧

在 3 週內實現高達萬倍的運算效能提升！NCHC、NVIDIA、OpenACC「NCHC Open Hackathon」黑客松，提供開發者實現 AI 創新最佳平台

NVIDIA 2025-03-13

分享本文

IDC 預測，2025 年全球 AI 支出將達 227 的 AI 經濟。當 AI 應用的需求正在快速成升 AI 發展的關鍵動能。面對這樣的趨勢 OpenACC 共同主辦「NCHC Open Hack 活動中，不僅匯聚來自全台各地的精英團的最佳平台。

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Podcast | Youtube | 聯絡我們 | 活力徵才 搜尋...

最新文章 AI 人工智慧 AI 機器人 智慧製造 新零售 金融科技 資安 主題特展 TO Highlight 全新一週

AI 人工智慧 雲端與運算

國網中心及 NVIDIA、OpenACC 舉辦「2025 NCHC Open Hackathon」黑客松，與台灣卓越研究團隊探索新世代 GPU 技術大幅加速強最新 AI 、量子運算等研究的嶄新技術創新

NVIDIA 2026-01-22

分享本文

熱門文章

1. 什麼是「上下文炸彈」？防守方第一次把 AI 提示注入反過來當武器
2. 為什麼 AI 做的 App 一眼就被看穿？三大跡象揭開 Vibe Coding 的 UI/UX 盲點
3. 機器人模擬資料不夠逼真？MIT 用 3 個 AI Agent 直接「生」出來
4. OpenAI Codex 負責人來台：下一代 AI 的定義，不再只由模型決定
5. 公開人形機器人 25 自由度機器手！1X：手是通往物理世界的 API

追蹤我們

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TO 會員電子報

還在靠社群媒體追新聞嗎？馬上訂閱電子報，最新科技趨勢、熱門活動不漏接！

國家實驗研究院 國家高速網路與計算中心 National Center for High-performance Computing

::: | EN | A- A A+

核心服務 創新技術 科研成果 動態資訊 關於我們

學研成果

日期 開始日期 ~ 結束日期 ex：2019/01/01 關鍵字 OpenACC 送出

科研成果

- 年度成果
- 學研成果
- 產業及新創合作
- 「御守臺灣・科技抗疫」專案

2026.01.14

【2025 NCHC, NVIDIA, OpenACC 黑客松】- 電腦視覺：無人機，運動科學，影像修復

2026.01.14

【2025 NCHC, NVIDIA, OpenACC 黑客松】- 大氣科學：加速運算，量子算法，人工智慧

2026.01.14

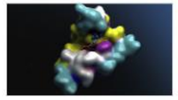
【2025 NCHC, NVIDIA, OpenACC 黑客松】- 前瞻創新：語言模型，量子化學，電漿模擬

2025.02.06

【2024 NCHC, NVIDIA, OpenACC 黑客松】— 生成式AI加速成果

2025.02.06

【2024 NCHC, NVIDIA, OpenACC 黑客松】— 人工智慧應用成果



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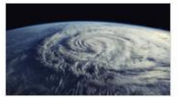
【2024 NCHC, NVIDIA, OpenACC 黑客松】— 醫療與生命科學成果

2025.02.06

【2024 NCHC, NVIDIA, OpenACC 黑客松】— 加速運算技術

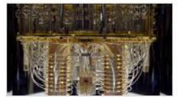


2025.02.06

【2024 NCHC, NVIDIA, OpenACC 黑客松】— 大氣科學模擬成果

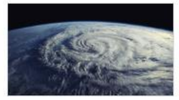
2025.02.06

【2024 NCHC, NVIDIA, OpenACC 黑客松】— 量子算法模擬成果



2024.01.30

【2023 NCHC, NVIDIA, OpenACC 黑客松】- HPC、DPU及量子運算加速成果



2024.01.30

【2023 NCHC, NVIDIA, OpenACC 黑客松】-大氣科學應用加速成果

2024.01.30

【2023 NCHC, NVIDIA, OpenACC 黑客松】-人工智慧應用加速成果

OpenACC More Science, Less Programming

OPEN HACKATHONS

國家實驗研究院 國家高速網路與計算中心 National Center for High-performance Computing

NVIDIA

2026 NCHC Open Hackathon

Important Dates

- 09 / 30 (Wed)：Registration deadline
  - https://www.openhackathons.org/s/siteevent/a0CUP00004AB2oc2AD/se000481
- 10 / 07：Connecting Mentors and Teams
  - Slack Channel, Email Group, Line Group ...etc.
- 10 / 14 (Wed)：Day 0: Kick-off Meeting 15:00~17:00 PM (Online)
- 10 / 21 (Wed)：Day 1: Scrum #1 Meeting 15:00~16:00 PM (Online)
- 10 / 28 (Wed)：Day 2: Scrum #2 Meeting 15:00~16:00 PM (Online)
- 11 / 04 (Wed)：Final Day 10:00~16:00 PM (In-person at NCHC)

Register Now!

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Thank You
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